PF-CT Calculation Package for PE Review — rev L4.2 (28 Oct 2026)

SUPERSEDED-FIGURES NOTICE (2026-11-02 cleanup). This rev L4.2 package (28 Oct 2026) is kept as the working record. Several figures in it are superseded: current masses are in data/mass_ledger.json (loco 20,404.2 kg pending freeze; tender 8,993.3 kg TB-3.15e; coaches F7 r5b 13,414.37 / 10,693.71 kg loaded), and current rulings are on status.html (sand system DELETED - sand dome is cosmetic trim only; 60 lb bolted relay rail; Loco No. 27; 4 coaches; C4 wheelchair boarding; level layout F; T04 spring brake is parking/holding only; PX-1 = ASCO 8327G041 in heated box; DF1 bogie on HOLD, baseline F7 r5b on L3 trucks; 6 mph governed operating limit, 6.5 mph HMI warning, 7 mph independent trip, 3 mph max if cameras fail; dead-loco tow ≤ 1 mph). Rows marked [SAND DELETED] are historical and must be re-run without sand. Every row is a desk calculation; NOTHING has been physically tested. Not PE-reviewed.

Pulsipher Farms Christmas Train · 36 in gauge diesel-hydraulic locomotive (8 × Ø1,450 drivers), tender T2, 4 coaches R2 · Logandale, Nevada
Status: PRELIMINARY - prepared by the owner team for review by a Nevada-registered PE. Not for construction until stamped. Supplier values not in hand are marked AWAIT; engineering choices not fixed by a code are marked ASSUMED for the PE to confirm or change.

PROFESSIONAL ENGINEER - REVIEW / STAMP
PE stamp and signature

Name: _____
NV PE No.: _____
Date: _____
Disposition: ☐ Approved ☐ Approved as noted ☐ Revise and resubmit

Numbered index

No.	Section	Calcs	Max util (excl. spec-requirement rows)
0	Design basis, masses, criteria	-	-
1	Locomotive frame and body	21	0.95
2	Trucks and suspension (powered, leading, tender/coach)	9	0.53
3	Axles and bearings	10	0.97
4	Drive (rev L4.1 Poclain MS05 2-speed + ANSI 120-2 chain 3:1, governed 6 mph) and hydraulics	87	1.00
5	Brakes - service, emergency, parking	17	0.90
6	Couplers, drawbars and end restraint	25	0.96
7	Tender T2 - structure, engine/pump/genset mounts, fuel, cooling	8	0.94
8	Generator sizing and selection recommendation	5	0.90
9	Coaches R2 - frame, body, roof, floor, doors, stairs, seating, restraint, egress	27	0.71
10	Electrical (train supply) and coach HVAC	4	0.92
11	Wind - 40 mph no-run, overturning, tie-downs	11	0.86
12	Track / rail interface - wheel profile, gauge, curve clearances	33	1.00
A	Fastener torque schedule	-	-
B	Weld schedule	-	-
C	Fix log (failures found and how they were fixed)	-	-

No.	Section	Calcs	Max util (excl. spec-requirement rows)
D	Revision history	-	-

Total calculation rows: 257. Utilization = demand / capacity (for FS rows: required / available). Rows that failed as-found are kept, shaded, and marked with their fix. Rows with util = 1.00 and demand = capacity are specification-requirement rows (the minimum rating a supplier part must meet), not a check at its limit.

0 Design basis (summary - full text in PF-CT Design Basis)

• Train mass (loaded, T2 tender, L4 drive): 81,513 kg; W = 799.4 kN. Loco 19,934 kg (L4.2: safety-chain parts 32.2 kg (2 × PL44.5 A572-50 lugs faced to 42.5 in the jaw zone + PL16 diaphragms, 1-1/8 in G-2130 shackles, Crosby S-1325A couplers; X11150; replaces the PL40 10.2 kg), Ø120 traction pins + GE120ES-class housings +2 × 22 kg (X4075/7275), sanders on both trucks (rear boxes 46 kg body-mounted X6400 + 8 kg nozzles X7275, front upgrade 6 kg), float bypass block 14 kg, T04 orifices / 2-channel vent switch / 96-tooth pony wheel 7 kg, brake-release piping 38.5 kg GROSS (Loco Bot E7, replaces the 5.0 kg net), MR feed JL-A→BP-1 11 kg (X8190), 20 L release reservoir 14 kg; L4.1: + T04 release circuit rev (6.0 L + 1.0 L accumulators, hand pump, shuttles; steering 7) 20 kg X5675, + low traction pins 2 × 55 kg (X4075/7275, ASSUMED), Loco Bot close-out 21.3 kg (saddle gussets 12.7, SD-1 caps 3.2, SD-2 clamps 5.4), + HL-1 headlamp 25 kg (upper bound) X175, TC/valve package 28 kg X5675, ballast clamp bars 21.1 kg X4075, hub 31.4 kg/axle (Ø350 flange), Loco Bot ledger 01:57 (barrel 2,076.2, saddles 195.7, bell 14.3, whistle 4.9, slats 21.9, L-107 -1.0); no flow divider. Loco = RT1-corrected base 16,657 (INCLUDES the L3 3 mm hood 898 kg) + L4 drive delta + Loco Bot shell ledger rev 2 (Steering 17): L-108 barrel 2,280.6 incl. saddles (S4N) - 898 hood, L-109 smokebox 227.9 IN FULL (the L3 smokebox front left with the ~1,000 kg L3 cab/front - no double count), L-110 fittings 51, L-107 P1-L net +28.2 + SS± HSS3x3x1/8 +9.1, cab L-111 C1 813.9 @X9283 replacing the ~1,000 kg L3 cab/front, fit-out 6, FT-1 15.6, BT-1 battery 68.2 in the Y- front pocket (base also INCLUDES the L3 PL6 deck 1,248) + front ballast plates trimmed 1,056 -> 150.8 (2 x 75.4); pilot slats 21.9 and JL plates 9.7 (ledger rev 2) + pony own ballast 1,250; jumpers 40 kg in train mass only; L4 drive delta +571 kg, part ASSUMED); tender 8,706.5; coaches 3 × 13,200 + 11,000 loaded. [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
• Driver adhesion weight 171.7 kN (lead truck 23.8 kN). μ dry 0.25 / wet 0.15 / sanded 0.25 (ASSUMED rail practice). [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
• Speed: GOVERNED 6 mph maximum operating speed - OWNER-SELECTED LIMIT (28 Oct 2026), PE TO CONFIRM; hard limit = H1P053 PUMP stroke stop sealed at 42 cc + HARD tamper-sealed 2,600 rpm engine limit (L4.2: 5.96 mph loaded, 6.47 mph no-load high-idle check < 7 mph trip) + PLUS+1; overspeed trip 7 mph by a 2-channel speed relay INDEPENDENT of PLUS+1 → emergency brake + pump neutral. Hardware is capable of ~7.6 mph with the stop removed - margin only, not permitted. Brakes, curve and passenger checks use 6 mph (trip case 7 mph).
• design grade 2% GOVERNING (max 2.0% on tangents, compensated on curves - route redline 28 Oct 2026); min. radius 100 m (design basis; R50 under study, see §12); gauge 914.4.
• Steel ASD: 0.6Fy bending/tension, 0.4Fy shear at ×1.3 service; Fy at ×2.0 exceptional. Welds E70, 0.3·Fu throat.
• Wind: 40 mph gust no-run; parked 100 mph 3-s gust (ASCE 7-22, ASSUMED - PE to take site value).

1 Locomotive frame and body

Governing codes and standards

• AISC 360-22 (ASD) for members and connections; AWS D1.1/D1.1M:2020 welding
• ASTM F2291-23 §6 (design loads, patron areas), F1193 (quality/manufacture), F770 (operations)
• IBC 2021 §1607 for walking-surface, guard and grab loads (reference only - PE to confirm applicability)

Load cases and basis

• DL+equipment ×1.3 service vs 0.6Fy; ×2.0 overload (track irregularity / jacking) vs Fy
• Drawbar 70 kN service, 200 kN ultimate (pin/box/rail) - ASSUMED basis, see §6
• Ballast 3 g long. / 2 g vert. / 1 g lat. (ASSUMED, F2291 equipment restraint)
• Jacking: 2 × loco weight on 4 pads
• Cab roof 20 psf live + 300 lb point; treads 300 lb; grabs 200 lb

Hand calculations

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
C4	Loco main rail HSS12x6x3/8, vertical ×2.0 (uniform equivalent)	Sprung body 8439 kg (LT42 reactions) spread uniformly; overhang 3775 mm	see PF-CT_R2_calc_notes (calcs.py)	90.1	345.0	MPa	0.26	OK	Ballast sits directly over the front bolster, adds no frame moment.
C4b	Loco main rail at rear sill, drawbar 70 kN service + vertical ×1.3	Eccentric drawbar: rail centroid Z1302 − coupler Z550 = 752 mm; moment shared by both rails via end sill + knee gussets	see PF-CT_R2_calc_notes (calcs.py)	105.7	207.0	MPa	0.51	OK	
C4b	Loco main rail at rear sill, drawbar 200 kN ultimate + vertical ×1.3	Eccentric drawbar: rail centroid Z1302 − coupler Z550 = 752 mm; moment shared by both rails via end sill + knee gussets	see PF-CT_R2_calc_notes (calcs.py)	193.3	345.0	MPa	0.56	OK	
C8	Loco body bolster box (PL12 flanges 350, PL10 webs, 305 deep), rear, ×2.0	Rear body reaction 5873 kg (LT42) at centre plate, span 480 mm (rails Y±240, rev L1)	see PF-CT_R2_calc_notes (calcs.py)	9.7	250.0	MPa	0.04	OK	
C8f	Loco front bolster with full 1.5 t ballast, ×2.0	Ballast load lands on rails directly; conservative to put on bolster	see PF-CT_R2_calc_notes (calcs.py)	6.7	250.0	MPa	0.03	OK	

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
C9a	Ballast pocket end wall PL20, 3 g longitudinal	750 kg per side, resultant at mid-height of 255 mm stack; wall acts as cantilever 480 wide	see PF-CT_R2_calc_notes (calcs.py)	87.6	150.0	MPa	0.58	OK	3 g longitudinal / 2 g vertical / 1 g lateral ASSUMED (engineer to set per ASTM F2291).
C9b	(Original idea) PL12 end wall, 3 g longitudinal	Shows why the end walls are PL20	see PF-CT_R2_calc_notes (calcs.py)	243.3	150.0	MPa	1.62	FIXED - PL20 end walls (C9a)	
C9c	Ballast hold-down 4× M20 cl. 8.8 rods, 1 g up + preload	Each rod tensile stress area 245 mm², proof 600 MPa; ASD limit	see PF-CT_R2_calc_notes (calcs.py)	1,838.7	70,419.2	N	0.03	OK	
C10a	Cab step tread PL8 × 240, 500 span, 300 lb point	IBC 1607 stair tread 300 lb concentrated (ASSUMED code basis)	see PF-CT_R2_calc_notes (calcs.py)	65.1	150.0	MPa	0.43	OK	
C10b	Cab step tread deflection	Limit L/240	see PF-CT_R2_calc_notes (calcs.py)	1.7	2.1	mm	0.81	OK	
C10c	Grab iron Ø32 solid A36, 1,200 span, 200 lb any direction	OSHA 1910.29 / IBC 1607.9 200 lb (ASSUMED code basis)	see PF-CT_R2_calc_notes (calcs.py)	83.0	150.0	MPa	0.55	OK	
C10e	Cab roof purlin HSS2x2x1/8 @600, 20 psf live + dead	Span 2,700 mm simply supported	see PF-CT_R2_calc_notes (calcs.py)	77.6	207.0	MPa	0.38	OK	
C10f	Cab roof purlin HSS2x2x1/8 deflection	Limit L/180	see PF-CT_R2_calc_notes (calcs.py)	11.6	15.0	mm	0.77	OK	
C10g	Cab roof purlin HSS3x2x1/8 @600 deflection (adopted)	Limit L/180	see PF-CT_R2_calc_notes (calcs.py)	4.4	15.0	mm	0.29	OK	
C10h	Cab roof purlin HSS3x2x1/8 @600 bending (adopted)		see PF-CT_R2_calc_notes (calcs.py)	43.9	207.0	MPa	0.21	OK	
C16a	Cab clear headroom (deck to purlin underside) vs 6 ft 3 in crew + shoes	Roof top 3700 - 3 skin - 76.2 HSS3x2 purlin = 3621; cab floor PL6 top 1605; 6 mm floor mat	see PF-CT_R2_calc_notes (calcs.py)	1,905	2,009.8	mm	0.95	OK	1,905 mm standing target ASSUMED (owner/crew to confirm)
C16b	Route overhead clearance adopted vs train height + 300 sway/bounce/track-lift + 300 decorations	Adopted R2: nothing lower than 4,300 mm above top of rail within 2,000 mm of track CL (lights, wires, signs, branches, arches)	see PF-CT_R2_calc_notes (calcs.py)	4,300.0	4,300	mm ARL	1.00	OK	

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
L1-J	Jacking pad - bolster web local yielding (AISC J10.2), 2 webs PL10, lb 150	P = 2·19,934 kg·g/4 pads	Rn/Ω = 2·Fy·tw·(5k+lb)/1.5, k=12	97.743	700.0	kN	0.14	OK	
L1-H1	L-107 blind-bolt holes Ø26 in main-rail OUTER webs (24 holes, 6 stations) - fatigue stress range at the hole vs AISC App. 3 Cat D threshold	hole Z1360 (57.6 mm off the NA), Ix loss 0.93 %	Δσ = 2·0.3·σ_service(y_hole)	10.3	48.0	MPa	0.21	OK	APPROVED with conditions (edge/weld-toe rules, see Loco Bot review). C4 util with holes 0.262
L1-H2	L-107 RB-1 bracket: Hollo-Bolt HB16 per-bolt shear + tension, worst L7-B3 (Loco Bot bracket_load_table.csv, 3 g long.)	V 15.4 kN, T 8.9 kN per bolt (L7-B3 / L7-B2)	linear V/30 + T/21 ≤ 1 (Lindapter SWL, conservative)	0.51	1.0	interaction	0.51	OK	Web bearing 15.4 vs 72.6 kN (util 0.21); 28 of 30 holes approved, 2 end-sill holes rejected
L1-H3	L-107 SD-1 bolster hold-down HB16 in PL12 bolster top (4 holes) - uplift	5.99 kN per bolt	T/21 kN SWL	5.99	21.0	kN	0.28	OK	APPROVED if the holes move to Y±90 (≥ 50 mm from the rail-web weld toe). Optional 2 holes in the rear end sill (drawgear zone): REJECTED

Materials

- Main rails HSS12x6x3/8 ASTM A500 Gr C (Fy 345); sills, crossmembers HSS A500C
- Bolster boxes, gussets, plates ASTM A36; drop box PL16 ASTM A572 Gr 50
- Cab/hood: 10 ga (3.4) A1011 CS skin on HSS2x2x1/8 posts; roof purlins HSS3x2x1/8 @600 (C10g/h)

Welds (AWS D1.1, E70XX, prequalified WPS unless noted)

- Crossmembers to rails: 6 mm fillet all round (E70, D1.1 prequalified)
- Drop box to rails/sill: 10 mm fillet all round, 100% VT + MT (C5w)
- Bolster box seams: 6 mm fillet both sides; centre-plate ring 8 mm fillet all round

Bolts and torques

- Ballast hold-down 4× M20 cl. 8.8 rods, 441 N·m (dry K 0.20) - C9c
- Cab-to-frame: M16 cl. 8.8, 219 N·m (dry K 0.20), Nord-Lock washers

Fixes logged

- F13: ballast end wall PL12 → PL20 (C9b NG → C9a 0.58)
- F04/C16: cab roof 3,700, purlins HSS3x2x1/8 (deflection)

2 Trucks and suspension (powered, leading, tender/coach)

Governing codes and standards

• AISC 360-22 (ASD); AWS D1.1
• EN 13749:2021 load cases used as a method reference for truck frames (ASSUMED - PE to confirm)
• SMI Handbook of Spring Design / EN 13906-1 for coil springs

Load cases and basis

• Vertical: sprung load ×1.3 service, ×2.0 exceptional (EN 13749 style)
• Lateral 0.35 × axle load at flange
• MS05 motor bracket on the transom: chain pull at the 430 bar strength basis (HPRV 380 + spike) + motor reaction torque (§4)

Hand calculations

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
C3a	Coach truck transom 2×HSS10x4x1/4 upright, ×2.0	P=46.5 kN at centre plate, span 1449 mm between sideframes	see PF-CT_R2_calc_notes (calcs.py)	65.9	345.0	MPa	0.19	OK	
C3b	Coach truck sideframe PL25×170 A36, ×1.3	Simply supported on axlebox springs 1,200 mm apart, transom load at centre	see PF-CT_R2_calc_notes (calcs.py)	80.0	150.0	MPa	0.53	OK	
C3b	Coach truck sideframe PL25×170 A36, ×2.0	Simply supported on axlebox springs 1,200 mm apart, transom load at centre	see PF-CT_R2_calc_notes (calcs.py)	123.1	250.0	MPa	0.49	OK	
T1-a	Powered-truck inside sideframe (PL30 chords as truss-girder) bending, ×2.0	centre load 35.1 kN ×2.0, span 1,550 between springs; chords PL30×110 / PL30×80 at 520 c/c	σ = M·c/I, M = P L/4	19.572	250	MPa	0.08	OK	
T1-b	Powered-truck horn cheek PL30×(220 wide) shear from lateral 0.35×axle load	Y = 0.35·42.9 kN	τ = 1.5 Y / (2 cheeks·30·220)	1.707	100.0	MPa	0.02	OK	
T1-c	Transom PL16 box 300×200 + MS05 motor bracket (L4): weld 2×400×10 both sides, chain pull at 430 bar strength basis (HPRV 380 + spike) ×1.5 + reaction torque	F 54.7 kN, M 10.93 kN·m, T_react 3203 N·m	f = √(σ_b² + τ²) (weld, line method)	30.579	145.0	MPa	0.21	OK	Transom horizontal bending 7.5 MPa (negligible). E70, 0.3×483 = 145 MPa

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
T1-d	Primary spring envelope: coil spring solid height + travel vs 170 mm LT18 window	static 17.5 kN/box, rate 701 N/mm for 25 mm static; coil d26 D112 n5.1 (5160 hot-wound)	H_loaded ≥ (n+2)·d + 25 bump + 10 clash	219.6	170	mm	1.29	FIXED - rubber-metal chevron springs specified	FAILS for a coil: FIX - specify rubber-metal (chevron) primary springs, 1 per box, rate ~580 N/mm vert., ≥ 26 kN max, fits 170 mm (AWAIT supplier, e.g. LT35 quotes)
T1-e	Primary spring spec load: max (×1.8 dynamic) vs supplier rating to be met	17.5 kN × 1.8	F_max = 1.8 F_static	31.554	31.554	kN	1.00	OK	Specification row (util 1.0 by definition) - supplier must certify
T1-f	Traction-pin spherical plain bearing GE120ES-class (L4.2: the pin carries LONGITUDINAL load only → RADIAL rating) - required static rating at the 3 g truck case	F = 144.2 kN (3 g × A-101 truck 4,900 kg at the pin)	C0_req = 2.0 · F (s0 2)	288.4	288.4	kN	1.00	OK	AWAIT SKF GE120ES radial C0 ≥ value (do not use until confirmed); L4.1 GE100ES vertical-load row superseded - the centre pad carries the vertical load

Materials

• Sideframes PL30 A572 Gr 50 (raised from A36 for toughness/fatigue), transom PL16 A572-50
• Primary springs: rubber-metal chevron, NR compound, steel interleaves (supplier)
• Coach/tender trucks: sideframe PL25 A36, transom 2×HSS10x4x1/4 A500C (C3)

Welds (AWS D1.1, E70XX, prequalified WPS unless noted)

• Horn cheeks to chords: CJP groove (D1.1 prequal. TC-U4b) + 6 fillet cap, UT 100% (cyclic)
• Transom box: 8 mm fillet both sides; L4 motor bracket PL25: 2×400×10 fillet both sides + 2 PL16 gussets (T1-c), MT
• Grind toe of all cheek and transom welds (fatigue detail cat. C → B')

Bolts and torques

• Axlebox keeper bolts M20 cl.10.9, 610 N·m (dry K 0.20), Nord-Lock
• Pivot cap M24 cl.10.9, 1,055 N·m (dry K 0.20)
• MS05 slide plate 4× M20 cl.10.9 in 50 mm slots, 610 N·m (dry K 0.20), Nord-Lock, after setting sag with the 2× M20 jack screws

Fixes logged

• Primary coil springs do not fit the 170 mm window (T1-d) → rubber-metal chevron springs specified
• Sideframe material A36 → A572-50 for cyclic margin

3 Axles and bearings

Governing codes and standards

• EN 13103-1:2017 (powered axle) method used as screen; AAR M-101 for steel grade reference
• ISO 281:2007 bearing life; ISO 76 static rating
• Goodman fatigue screen (Shigley) for the Ø150 pressed sprocket seat - NO hardening credit, fretting Kf 2.0 (Stress Bot)
• Lamé thick-cylinder press fit, μ 0.12

Load cases and basis

• Vertical 1.25 Q, lateral 0.35 × axle load at flange radius (EN 13103 simplified)
• Torque at 430 bar strength basis (HPRV 380 + spike) through the 3:1 chain (no efficiency credit) - structural peak; heaviest axle (L3-as-is REAR-2 + L4) for the seat
• Hub force at the sprocket seat: chain pull + 3 g on the sprocket/hub mass
• Bearing life: 20,000 h at 7.5 mph, P = 1.3 × (Fr + 2.5 Fa) with the chain-pull reaction on the HEAVIEST axle (Stress Bot #13)

Hand calculations

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
C7	Loco (front driver, full ballast) axle wheel seat Ø140 bending (fatigue screen)	Q=23.4 kN/wheel, ×1.25 vertical, Y=0.35×axle load on flange at r=725; bearing arm 176 mm	see PF-CT_R2_calc_notes (calcs.py)	63.2	120.0	MPa	0.53	OK	Allowable 120 MPa = EN 13103 wheel-seat permissible for A1N-class steel used as a screen; axle steel 4140 Q&T or EA1N; engineer to run full EN 13103/AAR check.
C7	Coach axle wheel seat Ø110 bending (fatigue screen)	Q=16.2 kN/wheel, ×1.25 vertical, Y=0.35×axle load on flange at r=300; bearing arm 169 mm	see PF-CT_R2_calc_notes (calcs.py)	52.2	120.0	MPa	0.44	OK	Allowable 120 MPa = EN 13103 wheel-seat permissible for A1N-class steel used as a screen; axle steel 4140 Q&T or EA1N; engineer to run full EN 13103/AAR check.
X1-a	L4 driven axle Ø150 pressed sprocket seat - EN 13103 screen (bending + hub moment + torsion at 430 bar strength basis (HPRV 380 + spike), nominal von Mises)	heaviest axle 49.12 kN; M_EN 17.28 + hub 3.03 kN·m	$\sigma = 32M/\pi d^3$, $\tau = 16T/\pi d^3$	66.3	110	MPa	0.60	OK	110 MPa = EN 13103 permissible for press-fitted seats, EA4T class (ASSUMED for 34CrNiMo6 - PE to confirm)

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
X1-c	Ø150 pressed seat fatigue WITHOUT hardening credit, fretting Kf 2.0, heaviest axle (Stress Bot) - Goodman n required 1.5 vs available	M_seat 20.32 kN·m, T 5.91 kN·m	1/n = σa/Se + σm/Su; Se = 0.5Su·ka·kb·0.814/Kf	1.5	1.57	n req/avail	0.95	OK	Ø140 gives n 1.37 (NG) → Ø150 raised seat on a Ø130 body. No keyway
X1-d	Seat-to-body shoulder Ø150/Ø130, r13 fillet (Kf 1.36) - Goodman n required 1.5 vs available	σa 128.1 MPa, Se 207 MPa	1/n = σa/Se + σm/Su	1.5	1.54	n req/avail	0.97	OK	
X1-e	Press fit Ø150 × 110, 0.12 mm min interference, μ 0.12 - slip torque vs 430 bar strength basis (HPRV 380 + spike) axle torque	p_min 64.0 MPa	T = μ·p·π·d²·L/2	9.609	29.9	kN·m	0.32	OK	SF 3.11; shrink fit ≤ 200 °C; oil-injection removal
X1-f	Pressed hub 4140 Ø240 hoop stress at max interference 0.17 mm vs 0.6·Sy	p_max 85.3 MPa	σθ = p(1+c²)/(1-c²)	194.7	393.0	MPa	0.49	OK	
X1-g	45T split ring to the Ø350 hub flange (L4.1): 12 × M16 cl.8.8 FITTED DIN 609/7968, PCD 300 - shear per bolt at 430 bar strength basis (HPRV 380 + spike) × 1.5	T_HPRV/(0.15 m·12) × 1.5	Fv = 0.6·fub·A(Ø17)/γM2	8.01	87.2	kN	0.09	OK	Bearing util 0.021; friction capacity 22.1 kN·m at 0.75 proof preload (HPRV util 0.43); tighten 219 N·m dry (153 lubricated), Nord-Lock, re-torque at 10 h
X1-h	Ø350 hub flange: bolt-hole edge distance (Ø17 reamed at PCD 300) vs 1.25·d0	e = 175 - 150 = 25 mm	e ≥ 1.25 d0	21.25	25	mm	0.85	OK	
X1-b	Loco inside axlebox bearing - required dynamic rating C, HEAVIEST axle + chain-pull reaction	P = 58.5 kN; L 53.0 Mrev	C_req = P · L^(3/10)	192.4	192.4	kN	1.00	OK	Spec row: select bearing with C ≥ 190 kN (L4.1 basis; computed 192.4, earlier ≥ 169 superseded) - supplier confirmation HELD

Materials

• Driven axles (L4): 34CrNiMo6 (EN 10083-3) Q&T Rm ≥ 1,000 MPa or EA4T, UT per ASTM A388, MT after machining; Ø150 raised sprocket seat fine-turned/ground Ra 0.8, r13 relief + undercut at the hub edge, NO keyway, no hardening credit taken
• Coach/tender/lead axles Ø110 4140 Q&T
• Bearings: sealed, grease Li-complex EP2; temperature sensor on driven axleboxes (ASSUMED)
• Pressed hub AISI 4140 Q&T Ø240×110 with integral Ø350×20 flange (L4.1, 31.4 kg); 45T 120-2 split rim ring with bolting web (Source Bot rev 3) on 12 × M16 8.8 fitted DIN 609/7968, 219 N·m

Bolts and torques

• Axle end caps M16 cl.10.9, 3 per end, 313 N·m (dry K 0.20) + tab washer

Fixes logged

• Outboard Ø80 axle stub + DT5 key (C7k old NG, F03) eliminated (L2/L3); axle half-length 724 → 560
• L3 Ø120 keyed reducer seat REMOVED with the Dodge reducer (L4)
• L4: Ø150 pressed seat passes WITHOUT hardening credit on the heaviest axle (Stress Bot) - Ø140 would fail (n 1.37)

4 Drive (rev L4.1 Poclain MS05 2-speed + ANSI 120-2 chain 3:1, governed 6 mph) and hydraulics

Governing codes and standards

• Poclain MS05 catalog + Application Engineering review (radial load/offset, shaft code 5 DIN 5480 W55x3x17, shift on the move, rail traction duty)
• ANSI/ASME B29.1 roller chain; maker (Tsubaki) allowable-load and lubrication practice; ISO 4413 hydraulic fluid power; SAE J517 / ISO 18752 hoses
• ASTM F2291 §6.6 (power/drive, speed control), F1193

Load cases and basis

• Start at 350 bar (H1P053 L35 pressure limiter) in 1st (468 cc); HPRV 380 bar (L4.2); the ~430 bar spike = STRENGTH basis (no efficiency credit)
• Motor start η_hm 0.90 base, 0.80 pessimistic (AWAIT Poclain starting efficiency); chain 0.97
• Adhesion: μ 0.25 dry / 0.15 wet / 0.25 sanded. Equal-torque cap 4·μ·lightest axle (parallel loop) GOVERNS - L4.1: no flow divider (Jahns HTO-4-35-FI rejected, D1-c) [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
• Curve resistance Röckl 400/(R-20) N/kN; start need W·(0.0208 + grade + curve)
• GOVERNED 6 mph by the pump stroke stop; independent 7 mph trip; hardware max 7.6 mph
• Engine V3307 45.6 kW net continuous (40.5 kW at 43 °C); pump allowance 35 kW (34.5 kW with the compressor clutched)

Hand calculations

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
D1-a	Start 2% tangent - DRY, equal torque (L4.1 electronic traction control, no divider): need vs min(TE 350 bar, 4·μ·Q_min)	need 32.61 kN; TE 37.7 kN; lightest axle F2 37.11 kN	F = min(TE, 4·0.25·Q_min)	32.61	37.1	kN	0.88	OK	Margin 1.14 (rule ≥ 1.10 on 2% by front-ballast sizing, 2 × 75.4 kg plates)
D1-as	Start 2% tangent - WET, SANDED (μ 0.25), equal torque	sand MANDATORY on wet grades / curves < R100 (auto-sand on slip)	F = min(TE, 4·0.25·Q_min)	32.61	37.1	kN	0.88	OK	Unsanded wet margin 0.68 (NG on 2% - operating rule: sand) [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
D1-a2	Start 2% + R100 uncompensated - DRY, equal torque (L4.1 electronic traction control, no divider): need vs min(TE 350 bar, 4·μ·Q_min)	need 36.61 kN; TE 37.7 kN; lightest axle F2 37.11 kN	F = min(TE, 4·0.25·Q_min)	36.61	37.1	kN	0.99	OK	Margin 1.01 (rule ≥ 1.10 on 2% by front-ballast sizing, 2 × 75.4 kg plates)
D1-a2s	Start 2% + R100 uncompensated - WET, SANDED (μ 0.25), equal torque	sand MANDATORY on wet grades / curves < R100 (auto-sand on slip)	F = min(TE, 4·0.25·Q_min)	36.61	37.1	kN	0.99	OK	Unsanded wet margin 0.61 (NG on 2% - operating rule: sand) [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
D1-a3	Start 0% + R50 - DRY, equal torque (L4.1 electronic traction control, no divider): need vs min(TE 350 bar, 4·μ·Q_min)	need 27.29 kN; TE 37.7 kN; lightest axle F2 37.11 kN	F = min(TE, 4·0.25·Q_min)	27.29	37.1	kN	0.74	OK	Margin 1.36 (rule ≥ 1.10 on 2% by front-ballast sizing, 2 × 75.4 kg plates)
D1-a3s	Start 0% + R50 - WET, SANDED (μ 0.25), equal torque	sand MANDATORY on wet grades / curves < R100 (auto-sand on slip)	F = min(TE, 4·0.25·Q_min)	27.29	37.1	kN	0.74	OK	Unsanded wet margin 0.82 (NG on 2% - operating rule: sand) [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
D1-b	2% start - pessimistic η_hm 0.80 (pending Poclain starting efficiency)	same	F = 4·p·V/(2π)·0.80·0.97·i/r	32.61	33.5	kN	0.97	OK	341 bar needed at η 0.80 vs 350 limiter
D1-c	REJECTED OPTION (L4.1): Jahns HTO-4-35-FI gear flow divider - TE with motor Δp capped by the 320 bar section rating vs 2% need	motor Δp ≤ 290.0 bar; 17.4 L/min/section at 2 mph vs min 12-16	F = 4·Δp·V/(2π)·η·i/r	32.61	31.2	kN	1.04	REJECTED OPTION - gear flow divider not fitted; electronic traction control (D1-m/m2)	Also below its minimum flow at every start, +160 kg, +heat → NOT FITTED; electronic traction control adopted (D1-m)
D1-c2	Divider at a start: flow per section at 0.5 mph in 1st vs catalog minimum 12 L/min (accuracy not defined below)	4 × 468 cc × n / ηv / 4	Q = V·n/ηv	12.0	4.4	L/min (min req/avail)	2.73	REJECTED OPTION - divider below its minimum flow at starts	Rejected-option row: the divider is not proven to lock the axles at start
D1-d	Single-wheelset spin at the HARD 2,600 rpm engine limit and the 42 cc stop (L4.2 high-idle check): standstill start in 1st (MS05 fail-to-1st; 2nd only through the PNOZ s30 ≥ 1.5 mph contact) vs MS05 HighFlow 1st max	one axle takes the full pump flow, no-load η_v 0.99	n = Q·ηv/Vg	229	370	rpm	0.62	OK	Rolling in 2nd (other axles turning): 378 rpm vs 430 (util 0.879). HighFlow valving REQUIRED (Classic 275 NG); traction control destrokes

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
D1-m	Electronic traction control (L4.1, replaces the divider): slip threshold vs ground-speed reference (3rd pickup on the unbraked pony)	MS05 sensor per motor + pony reference	spec	1.1	1.1	ratio	1.00	OK	Spec row: axle > v_ref × 1.10 or + 0.5 mph or > 0.5 m/s² → auto-sand, destroke to v_ref × 1.05 in ≤ 0.3 s (AWAIT Danfoss), ramp back; hill-start hold; rollback alarm. PLUS+1 software HELD vendor ask [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
D1-m2	2% start with traction control - DRY/SANDED equal-torque margin vs front-ballast rule 1.10	lightest axle 37.11 kN, 2 × 75.4 kg plates	4·0.25·Q_min / need	1.1	1.14	margin (rule/avail)	0.96	OK	Wet unsanded 0.68 → sand mandatory (unchanged rule) [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
D1-u	H1P053 input shaft G1 (14T 12/24) max torque vs V3307 engine peak 265 N·m (L4.1)	continuous 164.1 N·m (35 kW at 2,200 + charge)	T_peak / T_max	265.0	592.0	N·m	0.45	OK	AIE 'SAE B 13T' drive mates only G4 → engine coupling hub must be 14T 12/24 (AIE confirmation HELD); alternate G5 15T (370 N·m)
D1-u2	REJECTED OPTION: shaft G4 (13T 16/32, mates the AIE drive as quoted) vs engine peak 265 N·m	Danfoss rated 180 / max 222 N·m	T_peak / T_max	265.0	222.0	N·m	1.19	REJECTED OPTION - G4 shaft; G1 kept (D1-u)	G4 fails (1.19) → not used
D1-e	Pump H1P053 UNLIMITED input torque at the 430 bar STRENGTH basis (HPRV 380 + spike, L4.2) vs G1 input shaft rating 534 N·m	430 bar × 53.8 cc/(2π·0.93)	T = Δp·Vg/(2π·η_hm)	395.9	534	N·m	0.74	OK	35 kW cap (128.5 N·m) = anti-stall SOFTWARE; 34.5 kW while the compressor clutch is engaged
D1-f	6 mph GOVERNOR (L4.2): loaded speed in 2nd at the 42 cc stroke stop and the hard tamper-sealed 2,600 rpm engine limit	4 × 234 cc, 15:45, Ø1,450	v = Q·η/(Vg·i)	5.96	6.0	mph	0.99	OK	No-load at η_v 0.99: 6.47 mph (< 7 mph independent trip). Stop sealed at 42 cc; engine limit sealed by AIE (data HELD)

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
D1-d2	INFORMATION (double failure, not a design case): standstill spin in 2nd at 2,600 rpm / 42 cc, no-load, only if PLUS+1 AND the PNOZ s30 ≥ 1.5 mph contact both fail	one axle, η_v 0.99	n = Q·ηv/Vg	457	430	rpm	1.06	INFORMATION - double failure (PLUS+1 AND PNOZ s30); not a design case	INFO ONLY (util 1.064) - any single failure leaves the MS05 in 1st (fail-to-1st valves)
D1-n	Overspeed protection: independent 2-channel speed relay (SIL 2 / PL d class) vs governed 6 mph + 1 mph	2 dedicated sensors on different trucks	spec	7.0	7.0	mph	1.00	OK	Hard-wired to the BP exhaust valve + pump neutral (CCO), NOT through PLUS+1 (Stress Bot); tested daily (F770)
D1-o	MS05 pressure: HPRV 380 bar (L4.2 pressure revision, PL 350 unchanged, 30 bar spacing) vs motor max 450 bar			380.0	450	bar	0.84	OK	
D1-g	Main loop hose -16 (25.4 bore) velocity at the stop flow	Q 103.7 L/min	v = Q/A	3.411	4.6	m/s	0.74	OK	L4.2: -16 loop hose, code 62 / 24° S25 ends, whip restraints ≥ 420 bar WORKING (maker 420), ISO 18752 / SAE J517 impulse rating (ISO 18752 / ISO 6605, 133 % of WP) covering the ~430 bar transient; strength cases checked at 430 bar; Danfoss overshoot = information only
D1-h	Branch hose -10 (15.9 bore) per motor	Q 25.9 L/min	v = Q/A	2.176	4.6	m/s	0.47	OK	
D1-i	Hydraulic oil cooler duty vs SPEC ≥ 14 kW at 43 °C OUTSIDE air (tender left wall, own louvre)	35 kW, η_pump × η_motor × chain	Q = (1-η)·P	9.0	14.0	kW	0.64	OK	AWAIT cooler supplier curve at 43 °C; electric fan from the genset charger

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
D1-t	Hot-day engine power: traction allowance + fan/aux with the compressor clutched vs V3307 net at 43 °C	compressor off 37.3 kW	demand / P_43C	40.5	40.5	kW	1.00	OK	compressor electric clutch engages only at low traction demand or MR < 7.0 bar; anti-stall map drops to 34.5 kW while clutched; hydraulic cooler fan electric 24 V from the genset charger (not on the engine/alternator)
D1-j	2% hold speed (power-limited, best displacement, dp ≤ 350; 2nd only ≤ 300 bar) vs 2 mph target (ASSUMED)	35 kW × 0.855 × 0.92 × 0.95 × chain 0.97	v = P_rim/R	2.0	2.69	mph	0.74	OK	Hot day 2.65 mph
D1-qA	Route Alt A: worst START anywhere on the loop incl. curve resistance (Röckl 400/(R-20) N/kN, train-length averaged) vs adhesion-limited TE (L4.1 equal-torque dry cap 4·0.25·Q_min)	equivalent grade 2.0% (AREMA 0.8 lb/ton/° cross-check 2.199% → 33.06 kN)	F = W·(0.0208 + i + w_c) (× W_L4.1/W_RT1)	32.61	37.11	kN	0.88	OK	Wet: sand μ_req 0.208 (≤ 0.25 sanded). Hold incl. curves 2.72 mph (AREMA 2.52) vs 2 mph - curve compensation in the redline keeps grade + curve ≤ 2.0% [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
D1-qB	Route Alt B: worst START anywhere on the loop incl. curve resistance (Röckl 400/(R-20) N/kN, train-length averaged) vs adhesion-limited TE (L4.1 equal-torque dry cap 4·0.25·Q_min)	equivalent grade 2.0% (AREMA 0.8 lb/ton/° cross-check 2.199% → 33.06 kN)	F = W·(0.0208 + i + w_c) (× W_L4.1/W_RT1)	32.61	37.11	kN	0.88	OK	Wet: sand μ_req 0.208 (≤ 0.25 sanded). Hold incl. curves 2.72 mph (AREMA 2.52) vs 2 mph - curve compensation in the redline keeps grade + curve ≤ 2.0% [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
D1-qC	Route Alt C: worst START anywhere on the loop incl. curve resistance (Röckl 400/(R-20) N/kN, train-length averaged) vs adhesion-limited TE (L4.1 equal-torque dry cap 4·0.25·Q_min)	equivalent grade 2.0% (AREMA 0.8 lb/ton/° cross-check 2.185% → 32.96 kN)	F = W·(0.0208 + i + w_c) (× W_L4.1/W_RT1)	32.61	37.11	kN	0.88	OK	Wet: sand μ_req 0.208 (≤ 0.25 sanded). Hold incl. curves 2.72 mph (AREMA 2.53) vs 2 mph - curve compensation in the redline keeps grade + curve ≤ 2.0% [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
D1-qD	Route Alt D: worst START anywhere on the loop incl. curve resistance (Röckl 400/(R-20) N/kN, train-length averaged) vs adhesion-limited TE (L4.1 equal-torque dry cap 4·0.25·Q_min)	equivalent grade 2.0% (AREMA 0.8 lb/ton/° cross-check 2.199% → 33.06 kN)	F = W·(0.0208 + i + w_c) (× W_L4.1/W_RT1)	32.61	37.11	kN	0.88	OK	Wet: sand μ_req 0.208 (≤ 0.25 sanded). Hold incl. curves 2.72 mph (AREMA 2.52) vs 2 mph - curve compensation in the redline keeps grade + curve ≤ 2.0% [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
D1-r	R50 (Alt C intent) start on 2% WITH curve compensation 1.33% (grade on curve 0.67%) vs adhesion-limited TE	w_c(R50) Röckl 13.33‰, AREMA 13.97‰	F = W·(0.0208 + i + w_c) (× W_L4.1/W_RT1)	32.61	37.11	kN	0.88	OK	UNCOMPENSATED R50 at 2.0% needs 43.3 kN > 37.1 kN adhesion (NG) - Alt C must compensate R50 curves by ≥ 1.33% (grade ≤ 0.67% on the curve)
D1-s	R100 start on 2% WITHOUT compensation (sensitivity, if the track crew does not compensate) vs adhesion-limited TE	w_c(R100) Röckl 5.0‰ (AREMA 6.99‰ → 36.93 kN)	F = W·(0.0208 + i + w_c), L4.1 W (drive_L4 case 2% + R100 uncompensated)	36.61	37.11	kN	0.99	OK	SENSITIVITY only (basis: curves compensated). Hold 2.27 mph. Recommend compensation 0.04%/° rule ≈ 0.5% on R100, 1.33% on R50, 1.6% on R45
D2-a	Chain Tsubaki RS120-2: pull at 430 bar strength basis (HPRV 380 + spike) vs maker allowable load	T_axle 9,609 N·m on PD 546.2 mm	F = 2T/PD	35.2	51.7	kN	0.68	OK	SF on ultimate 7.3 (HPRV), 7.9 (heaviest-axle slip)

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
D2-b	Chain: continuous 2% running pull vs allowable			13.3	51.7	kN	0.26	OK	
D2-c	Motor shaft code 5 (W55x3x17) radial load at 400 bar vs catalog dynamic radial (sprocket at offset ≤ 0)	chart p.20		35.2	60.0	kN	0.59	OK	AWAIT Poclain AE for offset and shift-on-the-move
D2-d	Motor bearing life at continuous 2% running vs 20,000 h at 6 mph	L ∝ F^-10/3 from 10 Mrev at 30 kN		127.2	152.0	Mrev	0.84	OK	
D2-e	Chain wear take-up: centre growth at 3% elongation vs slotted-mount slide	C 417.6 → 451.4 mm		33.8	37.0	mm	0.91	OK	Replace the chain at 2% (3% absolute)
D2-f	L4.1 corrected chain slack (dL = 2·span/C·dC, Stress Bot r2): axle bump to tight chain at sag 17 mm, chain line horizontal, vs the ±22 mm hard bump stop - NO IDLER	span 376.1 mm, dL/dC 1.801; 3 mm primary shim: 28.0 mm	z_tight from $\sqrt{(\sqrt{(C^2+z0^2)}+S/(dL/dC))^2 - C^2} - z0$	22.0	30.8	mm	0.71	OK	Re-shim procedure: wheel-wear height at the centre plate/side bearers only; primary re-shim ≤ 3 mm then motor-slide reset at ride height (chain line horizontal ±3). L4 row used dL = dC (wrong).
D2-g	Oil-bath case radial clearance at 3% wear + ±25 travel vs 15 mm minimum	case inner r 332.7 mm vs OD/2 284.5 + ride-out 8.2 + 25		15.0	15.0	mm	1.00	OK	X -336..C_new+158 from the axle, Z axle +/-336 (was -310..C+125, +/-305 - FAILED at 3 %: the 15T would have hit the end wall)
D2-g2	Oil-bath case MOTOR end clearance at 3% wear with the Martin 15T OD 202 (L4.1; case sized on ~206) vs 15 mm	OD 202.0 (L4 206.1)		15.0	17.1	mm	0.88	OK	case unchanged; 15T to sideframe chord 85 mm (lateral)

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
D4-ri	L4.2 trip run-in (Stress Bot r3): worst loco-tender buff over CCO overspeed / 24 V loss (≥ 3 s swash ramp, MS05 fail-to-1st), FLOAT entry with the bypass failed closed, accidental T04 and a T04 line burst downstream of the choke (Ø1.2 port orifices, 2-channel vent switch) vs the R80 loco-LT allowance	Stress Bot s4 lumped-mass model, k 20/100 kN/mm, 1.0 mm play; tender TB-3.13d 8,706.5 kg, CG [2422.7, 11.3, 1110.4], brake 5.845 kN (0.671 m/s²); hydraulic trip force at 400 bar kept (above the HPRV 380 setting; the ~430 bar spike is a ms-scale transient); L4 as-was 67.0 kN	max buff over all design cases	25.3	33.2	kN	0.76	OK	Double failures (information only): 24 V loss + bypass failed closed 29.7 kN; T04 burst + both vent channels failed 34.3 kN. Direction change only at zero speed + ≥ 1 s ramp.
D4-wt	L4.2 forward 2 % start, sand on BOTH trucks (every driven axle, direction of travel), Ø120 traction pin (h 0.65 m) + laminated elastomer centre pad / PTFE yaw slide (pad share 0.083, μ ≤ 0.08) - margin rule ≥ 1.10	worn Ø1430 margin 1.112; degraded (trailing nozzles failed, carry-over μ 0.20) 1.101 new / 1.083 worn; rigid steel plate at μ 0.08 would give 1.079	4·μ·min(axle with transfer) / need	0.899	0.909	need/cap	0.99	OK	margin 1.112 (worn, governing) vs 1.10 [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
D4-wr	L4.2 reverse full-train start, sand on both trucks: OPERATING RULE grade ≤ 1.2 % (from the calc: 1.10 margin at 1.39 % new / 1.35 % worn; reverse pull cap of record 28.9 kN with 1.10: 1.21 % - GOVERNS)	reverse 2 % margin 0.899 (< 1.10: the lightest axle F2 is already sanded) - covered by the rule; PLUS+1 reverse pull cap 28.9 kN (supersedes 29.4)	g_rule ≤ min(g(1.10, worn), g(1.10, 28.9 kN cap))	1.2	1.21	%	0.99	OK	Track is LEVEL (Layout F, 0.0 %, Rail Bot §5b FINAL); 2 % checked as the conservative vehicle case [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
D4-hf	L4.1 hub shrink fit 0.15-0.20 (oil pressing deleted): slip torque, hub 30 K hot, 9.6 μm smoothing, removal groove deducted vs 1.5 × HPRV axle torque	δ_eff 0.0887 mm, μ 0.12	T = π d² L μ p / 2	14.397	16.7	kN·m	0.86	OK	SF at HPRV 1.74 (≥ 1.5); hoop 195 MPa < 393

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
D4-sp	L4.2 HIGH-IDLE CHECK: top speed at the HARD tamper-sealed 2,600 rpm engine limit with the 42 cc stop, no-load η_v 0.99 vs the 7 mph independent trip	2nd 234 cc, 15:45, Ø1,450; loaded 5.96 mph	v = Q·η/(Vg·i)	6.47	7.0	mph	0.92	OK	The L4.1 2,850 rpm high-idle case no longer exists (engine cannot exceed 2,600); AIE to set and seal (HELD)
D4-ch	L4.2 chain reversal with the zero-speed interlock + ≥ 1 s ramp (impact ~1.3) at the 430 bar STRENGTH basis vs ANSI 120-2 allowable	Stress Bot r2 #3; pull at 430 bar 35.2 kN	F = 1.3·F_430	45.8	51.7	kN	0.89	OK	breaking-strength certificate: see D4-mb
D4-mb	L4.2 chain minimum breaking load at the 430 bar STRENGTH basis: same factor as the old ≥ 302.5 kN at HPRV 400 → required MBL vs the SPECIFIED certified MBL (ANSI 120-2, mill certificate per lot)	ANSI B29.1 duplex 302.5 kN at HPRV 400 (pull 32.7 kN) -> factor 9.24; pull at 430 bar 35.2 kN	MBL_req = factor · pull_430	325.2	326	kN	1.00	OK	SPEC ≥ 326 kN certified. Standard Tsubaki RS120-2 (296 kN) does NOT meet it at 430 (util 1.099, see D4-mb2; it passes at HPRV 380: 287.4 kN, util 0.971) - higher-strength 120-2 = VENDOR BATCH, or PE acceptance of SF 8.41 on ultimate at the spike
D4-mb2	INFORMATION: standard Tsubaki RS120-2 published 296 kN vs the same-factor MBL at the 430 bar strength basis			325.2	296.0	kN	1.10	INFORMATION - standard 296 kN chain not accepted; SPEC certified MBL (D4-mb)	NOT ACCEPTED as standard - see D4-mb (passes at HPRV 380, 287.4 kN)
L41-sw	Loco Bot (c): L-108 saddle web PL10 out-of-plane at 3 g longitudinal WITH the 100×70 rivnut windows - FIX: 8 × PL10 45×150 gussets per saddle at Y±535/±685	as detailed 1101 MPa (NG, util 5.9)	σ = F·h / S	52	187	MPa	0.28	OK	+12.7 kg (6 saddles) in the ledger
L41-s1	Loco Bot (e): SD-1 bolster cap re-cut inboard to Y±50, HB16 at Y±90 (edge 40 ≥ 1.5 d0), cap bending under 8.05 kN uplift (LRFD)	PL10 × 200, lever 73.8	1.6 M / φ Fy Z	0.95	1.125	kN·m	0.84	OK	

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
L41-s2	Loco Bot (d): SD-2 end-sill hold-down by 4 clamp plates PL16 80×120 under the sill flange (no sill holes) - clamp plate bending	uplift 8.05 kN, 2 clamps/side	σ = M/S	17.7	150	MPa	0.12	OK	M16 8.8 4.03 kN vs 62.8 kN
L41-cb	Loco Bot (f): ballast clamp bars FB100×12×560 for 1 plate (75.4 kg) per side pocket - 2 g uplift + 3 kN rattle preload; plate located by the pocket walls	4 bars, 2 per plate	M = F/4·(L − b/2)	110	187	MPa	0.59	OK	rod nuts 30 N·m + Nord-Lock (not 441)
L41-w1	Ø1430 driver condemning limit - governed speed (42 cc stop, sealed 2,600 rpm) vs 6 mph	v ∝ D: 5.96 × 1430/1450	v ≤ 6.0	5.88	6.0	mph	0.98	OK	trip set in axle rpm for Ø1450 → 6.90 mph at Ø1430 (safe side)
L41-w2	Steering 7 (b): 2 % start at Ø1430 - forward equal-torque cap with transfer (worn tread −341 kg) vs need	margin 1.128	need / cap	0.887	0.909	-	0.97	OK	margin ≥ 1.10 basis; TE at 350 bar 38.2 kN
L41-w3	Reverse full-train start at Ø1430 (L4.2, sand both trucks, centre pad) - rule grade vs worn-wheel 1.10 grade	worn grade for 1.10: 1.35 %	g_rule ≤ g_cap	1.2	1.35	%	0.89	OK	operating rule 1.2 % (governed by the 28.9 kN reverse pull cap; L4.1 1.55 superseded) [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
L41-w4	L4.2 FINAL flange-height limit (Loco Bot E7 / Rail Bot §5b): WITHDRAW at 28 mm train-wide, re-profile ≤ 27; joint rail-head vertical wear ≤ 2 mm (rail height condemn ≤ 86.9 mm at bolt positions) - worn flange vs ASCE 40 joint-bolt head top	head top 32.5 mm below TOR new − 2 mm wear = 30.5; 30 mm limit WITHDRAWN; 6 mm joint limit WITHDRAWN (applies only > 1 m from rail ends)	h_f ≤ head top − joint wear	28.0	30.5	mm	0.92	OK	clear 2.5 mm at both limits; bolt protrusion ≤ 10 mm measured at install (worst 9.1 with flat bars ≤ 12.0 + 3/4 in B18.10 oval-neck H ≤ 12.7); pony 22 mm to the flangeway; frog flangeway ≥ 36 mm

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
L41-w5	Steering 7 (b): lower outline at Ø1430 (all truck items −10 mm)	min 255 → 245 ATOR	150 ≤ clear	150	245	mm	0.61	OK	
L41-t1	L4.2 T04 apply delay (choke Ø2.4 + 1.0 L damping accumulator; Ø1.2 orifice in each T04 port) - required ≥ 1.0 s for the run-in result	V 0.18 L at Δp 17.5 bar	t = V / Q_orifice	1.0	1.01	s	0.99	OK	rise 4.21 s ≥ 2.0; release 6.20 s vs the separate T04 target 8 s; verify by timed commissioning test
L41-t2	Steering 7 (a): manual-release accumulator - required V0 for 2 releases (2 × 0.58 L, 22 → 15 bar, pre-charge 13.5, adiabatic) vs 6.0 L	T04 volume 4 × 0.10 L ASSUMED	V0 = ΔV / [(p0/p1)^(1/n) − (p0/p2)^(1/n)]	5.45	6.0	L	0.91	OK	
L41-b1	Steering 8/9: LOCO brake release time WORST CASE (8 × 1.0 L calipers, 0 → 5.1 bar g, supply drooped to 5.6 bar, isothermal, + 1.0 s signal/relay ASSUMED) at the specified min total conductance C 3.0	Loco Bot 3.7 s at C 2.0 (1.8-7.3); this model 3.68 s fill at C 2.0	ISO 6358 fill	3.46	4.0	s	0.86	OK	C required ≥ 2.50; design budget C 5.26 → 2.40 s; commissioning timed test
L41-b2	Steering 8/9: release-path total sonic conductance - spec minimum vs parts budget (regulator 10 + relay 8 + 3/4 in trunk 8 in series, 4 × 1/2 in branches C 2.0 in parallel)	C_tot = (ΣC_i^-3)^(-1/3)	C_spec ≤ C_budget	3.0	5.26	dm³/(s·bar)	0.57	OK	Loco Bot line lengths / fittings to confirm trunk and branch C
L41-b3	Steering 8/9: MR after one loco release from cut-in (7.5 − 0.98 bar) vs regulator set 6.0 + 0.4 bar regulating head	48 L free air per release; recharge 17.8 s at 14 cfm	p_MR ≥ p_set + 0.4	6.4	6.52	bar	0.98	OK	
L41-b4	Steering 8/9: train steady brake-air demand (14.0 L calipers; 84 L free air per release) vs IR 2340 compressor 14 cfm	3.55 cfm steady	demand / FAD	3.55	14.0	cfm	0.25	OK	duty 25 %%

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
L41-b5	TB-3.13d tender brake (8,706.5 kg; calipers unchanged, 5.845 kN, mixed Coremo D 2N sets, pad wear limit 5.5 mm) vs loco 0.690 - decel mismatch vs 0.03 m/s² run-in allowance	tender 5.845/8.7001 = 0.672 / loco 0.690	abs(Δa)	0.018	0.03	m/s²	0.60	OK	tender softer: buff run-in covered by D4-ri
L41-e1	ERV roof pod: static allocation 3,450 ARL + kinematic allowance (bounce/ roll/lateral, ASSUMED 150 mm) vs K-401 overhead 4,300	static 3,450	z_kin ≤ 4,300	3,600	4,300	mm	0.84	OK	3,450 is a STATIC allocation; kinematic envelope checked against K-401
L42-dn	Descent 2 % (FLOAT mode: loop bypass + ≥ 3 s swash ramp) - FRICTION brakes sized for the full descent with the loaded train: descent force vs SERVICE friction force	train 81513 kg; shares loco 0.245 / tender 0.104 / coaches 0.651	F = m·g·(i - r)	15.99	39.35	kN	0.41	OK	Disc heat: continuous descent limit 2398 m (coaches govern, ΔT 250 K adiabatic) vs Layout F dead level; cooler in float 3.3 of 14 kW (util 0.24) - not overloaded
L42-dc	Float-mode cooler load vs cooler rating			3.3	14.0	kW	0.24	OK	bypass circulation only
L42-t1	T04 release time (normal, charge feed through the CDK 20 bar reducer, Ø2.4 choke + 1.0 L damping accumulator + Ø1.2 port orifices) vs the SEPARATE T04 release target	delay 1.01 s + rise 4.21 s	orifice/accumulator model, Cd 0.62	6.2	8.0	s	0.78	OK	MET. Air (friction) release ≤ 4 s is a separate target - L42-ar
L42-t2	T04 line burst DOWNSTREAM of the choke (hard lines / 4SH; Ø1.2 port orifice per motor): loco-tender run-in vs the R80 allowance	rise to 90 % in 3.32 s	Stress Bot s4	25.3	33.2	kN	0.76	OK	2-channel vent switch; accidental T04 (BP first) 23.8 kN

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
L42-t3	T04 charge-feed pressure at LOW IDLE (charge 21 bar min ASSUMED, CDK 20 bar) vs T04 full release 15 bar (ASSUMED)			15.0	20.0	bar	0.75	OK	margin 1.33; loop shuttle DELETED - no loop-pressure part in the T04 circuit
L42-t4	Tow / engine-off T04 release: 6.0 L bladder accumulator (pre-charge 13.5 bar) + hand pump - accumulator volume for 2 releases		Boyle	5.45	6.0	L	0.91	OK	97 hand-pump strokes for a full recharge
L42-p1	Traction pin Ø120 4140 Q&T - service: 1.3 × TE at the 430 bar strength basis (0.6 Fy)	lever 0.46 m ASSUMED	σ = M/Z	82	393	MPa	0.21	OK	GE120ES-class spherical plain bearing; +22 kg per pin vs Ø100/GE100ES (ledger)
L42-p2	Traction pin Ø120 - 3 g truck (A-101 4,900 kg, exceptional, Fy)		σ = M/Z	391	655	MPa	0.60	OK	pin in build/model.py (Z1110 → Z650 at X4075 / X7275)
L42-cp	Centre-plate friction in the start margin: laminated elastomer pad + PTFE yaw slide (pad share 0.083 ASSUMED, μ ≤ 0.08) - forward 2 % margin	k_pin 32.9 / k_pad 3.0 MN/m	4·μ·min axle / need	0.899	0.909	need/cap	0.99	OK	fwd margin 1.112 vs 1.10; pad stiffness and PTFE μ = VENDOR BATCH (ASSUMED)
L42-k2	K2-a OF RECORD (L4.2): max driver wheel at the PLUS+1 forward TE command limit 33.0 kN (1.01 × the 2 % need) incl. +42.8 kg final drive, Ø120 pins, P-106 pockets, gross brake piping/chains and the 59.8 kg NB-1 nose ballast at X450	reverse pull cap 28.9 kN → wheel 25.74 kN	truck-pitch + body transfer	25.89	26.0	kN	1.00	OK	loco 19934 kg; without NB-1 the start TE would load a wheel to 26.1 kN (Loco Bot drift) - FIXED by NB-1
L42-sf	Sanders on BOTH trucks (level switch, flow proof, hill-hold) - forward 2 % start margin, worn Ø1430			0.899	0.909	need/cap	0.99	OK	margin 1.112 [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
L42-pz	Pony reference: ~96-tooth target wheel - pulse rate at 0.5 mph (worn) vs the 10 Hz minimum for the 100 ms filter	2 Hall sensors in quadrature; agreement check vs axles; PNOZ s30 kept independent	f = v·z/(π·D)	10.0	10.5	Hz	0.95	OK	92 teeth minimum
L42-wd	Wind on curves: R80 dry at 25 mph wind (interim rule: 25 mph limit on R ≤ 80, no full-pull starts with a lubricator out) - L/V margin	40 mph dry 0.939 (covered by the rule); lubricated 40 mph 1.287		1.0	1.018	1/margin	0.98	OK	Layout F has no R70
L42-lg	Safety-chain lug PL44.5 A572-50 faced to 42.5 mm + 1-1/8 in G-2130 shackle - 1.5 × TE at the 430 bar strength basis vs shackle WLL 93.2 kN	69.4 kN		69.4	93.2	kN	0.74	OK	lug util 0.17; each chain = 2 × G-2130 + 2 × S-1325A + 1-2 links 5/8 G100; coupler WLL ≥ chain WLL PASS (Loco Bot E8)
L42-ar	Brake AIR release time (QEV option B kept, MR ≥ 5.6 bar) vs the ≤ 4 s air release target	C_spec_min 3.2 per branch; relay/regulator C ≥ 8-10; valve purchase spec C ≥ 0.31	ISO 6358 fill	2.36	4.0	s	0.59	OK	Loco Bot E7/E8; coach release indication dual-channel per Electro E3.6h (PL d Cat 3)
L42-mr	MR after one release from the 6.0 bar governor minimum (coupled MR ~190 L) vs the 5.6 bar used in the release time	release volume 10.26 L	Boyle	5.6	5.68	bar (min)	0.99	OK	20 L local reservoir DELETED
L42-qa	QEV option A (QEVs deleted): spring-apply exhaust time through the relay, worst branch, cold -10 °C, 6.0 → 0.5 bar g (ASSUMED full-apply pressure)	relay exhaust C 8 ASSUMED; to 0.3 bar g 2.38 s	ISO 6358 discharge	2.13	2.0	s	1.06	INFORMATION - QEV option A rejected; option B kept (L42-qb)	INFORMATION: option A FAILS (shortfall 0.13 s) although its 7 mph / 2 % stop passes (21.7 m vs 30) → option B KEPT
L42-qb	QEV option B (maker-certified ≥ 15 bar 1/2 in QEV, vendor batch): spring-apply time, worst branch, cold		ISO 6358 discharge	0.94	2.0	s	0.47	OK	7 mph trip stop on 2 % down with the apply delay: 20.1 m vs 30 m; piping 43.7 kg gross

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
L42-tv	Tow valve across loop A-B: HYDAC KHB3K DN06 class (500 bar) at the 430 bar strength basis (DN25 310 bar NOT allowed on the loop)	tow ≤ 2 mph in 1st: 67.1 L/min	p/rating	430.0	500.0	bar	0.86	OK	util 0.86
L42-tq	Tow valve DN06 sizing: Δp at the 2 mph tow flow (4 MS05 in 1st, worn Ø1430) vs 30 bar limit	bore 6 mm, K 1.5 ASSUMED	$\Delta p = K \cdot \rho \cdot v^2/2$	10.2	30.0	bar	0.34	OK	tow drag 1.27 kN, heat 1.14 kW; confirm with the HYDAC DN06 curve (vendor batch)
L42-hs	Loop hose Manuli DIAMONDSPIR H10018025 (ISO 18752-C, 560 bar WP) vs the 430 bar spike / strength basis	HPRV 380 → 0.679	p/WP	430.0	560.0	bar	0.77	OK	OD 41.2, 2.91 kg/m; impulse rating per ISO 18752-C
L42-hb	Loop hose minimum bend radius 350 mm vs the tender's tightest bend at R70 (394.3 mm, Tender Bot)			350.0	394.3	mm	0.89	OK	util 0.888
L42-gb	GRADE BASIS: track Layout F dead level (0.0 %, Rail Bot §5b FINAL); vehicles checked at 2 % as the conservative case - reverse full-train start limit (all files)			1.2	1.21	%	0.99	OK	governed by the 28.9 kN reverse pull cap with 1.10
D2-h	15T bore spline N55x3x17 on MS05 shaft code 5 (W55x3x17): flank pressure at the 430 bar STRENGTH basis × KA 1.5 - 15T made to order in 4140 Q&T 28-32 HRC (L4.2)	T 3,203 × 1.5 N·m; L 60 engaged	$p = 2T/(d \cdot z \cdot h \cdot L \cdot \varphi)$	102.6	150.0	MPa	0.68	OK	C1045 catalogue hub would be 1.03 (NG) - Martin D120B15 stock NOT accepted; REQUIRE ≥ 58 mm engaged, hub face to shoulder ≤ 2 mm, end plate. Hub torsion util 0.09

Materials

• Motor: Poclain MS05 468/234 2C HighFlow, T04 brake, shaft code 5, DIN 5480 W55x3x17 (male), speed sensor (model code AWAIT Poclain)
• Chain: ANSI 120-2 (Tsubaki RS120-2 or equal), 54 links, oil bath ISO VG 220 EP
• Sprockets: 15T Martin D120B15 (OD 202) hub broached DIN 5480 N55x3x17 (female), ≥ 58 mm engaged; 45T SPLIT rim ring with bolting web, 12× M16 8.8 fitted DIN 609/7968 (219 N·m) to the Ø350 flange of a 4140 pressed hub (31.4 kg) - replaceable in place, no pressing
• Oil-bath case 3 mm steel on the motor bracket, floating seal plate at the axle
• Hydraulic fluid ISO VG 46 HV; EXTERNAL charge filtration HYDAC DF 60 pressure filter with clogging indicator (grade ASSUMED β10 ≥ 200)

• Pump: Danfoss H1P053 with CCO (control cut-off, wired to the Pilz s30 7 mph trip), G1 shaft (14T 12/24; engine coupling hub 14T 12/24 on the SAE B pad), L35 (PL 350 / HPRV 380), stroke stop 42 cc sealed + hard 2,600 rpm engine limit, PLUS+1 (anti-stall 35 kW, shift, traction control)
• L4.1 NO flow divider. Traction control: 3rd speed pickup on the pony tone ring, loop A/B transducers, auto-sanders [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
• L4.1 loco valves (steering 7): T04 release from the loco loop = [L4.2: loop shuttle DELETED - T04 fed from the H1 charge circuit] + 10 µm filter + 3-way reducing/relieving valve 20 bar + 3/2 solenoid NORMALLY VENTED (HydraForce SV08/SV10-30 class) + choke Ø2.4 + 1.0 L damping accumulator + brake-line shuttle; manual release = check + 6.0 L bladder accumulator + 12 cm³ hand pump + guarded 3/2 valve; loss of loop/charge APPLIES the T04; no 4th dry-break; tow = lockable 2-way ball valve ≥ 420 bar (L4.2) -16 across A/B (HYDAC KHB class) with limit switch

Welds (AWS D1.1, E70XX, prequalified WPS unless noted)

• Motor bracket PL25 A572-50 to transom: 2 × 400 × 10 fillet both sides + 2 PL16 gussets (T1-c), MT

Bolts and torques

• MS05 mounting flange cl.10.9 per Poclain (AWAIT)
• Slide plate 4× M20 cl.10.9, 610 N·m; jack screws M20
• 45T ring web to the Ø350 hub flange 12× M16 cl.8.8 fitted DIN 609/7968 at PCD 300, 219 N·m dry, Nord-Lock, re-torque at 10 h (7.45 kN/bolt at HPRV × 1.5)
• Brake calipers M20 cl.10.9, 610 N·m (dry K 0.20)

Fixes logged

• L4: Dodge TA7315H15MS + H1B110 + Gates belt REMOVED (owner: no belt) → MS05 + RS120-2 chain 3:1, nothing outside the wheels
• Equal-torque start limited by the lightest axle → L4.1: gear flow divider REJECTED (below min flow at starts, 320 bar section rating < need, +160 kg) → electronic traction control + front ballast for ≥ 1.10 dry; sand mandatory on wet grades [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
• Pump shaft: AIE SAE B 13T drive would force G4 (fails 1.19 on engine peak) → keep G1 + 14T 12/24 coupling hub (D1-u)
• Spline engagement 70 → 60 mm (full code-5 spline): 95 MPa, util 0.95 (D2-h)
• HighFlow valving required for single-axle spin (D1-d)
• Overspeed trip made independent of PLUS+1 (D1-n)
• Cooler raised to ≥ 14 kW at 43 °C outside air (D1-i); compressor clutch gating + 34.5 kW anti-stall (D1-t)
• Chain case enlarged to ≥ 15 mm radial at 3% + travel (D2-g)
• Charge pump: Danfoss 10 % rule gives ~193 cc for LSHT motors (meaningless) → OPEN item; charge filter added

5 Brakes - service, emergency, parking

Governing codes and standards

• ASTM F2291-23 §6 (fail-safe brakes, controls) and F1193; ASTM F770 (operations)
• Air components to SAE J1410/J1402 & FMVSS 121 component practice (reference); NFPA 70 for controls
• 49 CFR 232 used as practice reference only (not applicable)

Load cases and basis

• Brake force set at μ_use 0.12 of TARE (empty) axle load on every braked axle → no wheel slide on wet rail empty or loaded (μ wet 0.15)
• Braked: 4 loco driven axles (axle disc, 2 calipers), 4 tender axles, 16 coach axles; lead truck unbraked
• Emergency = spring-applied on BP exhaust (fail-safe); service = 70% of emergency; parking = same spring units
• Cases on 2% DESCENDING, reaction/propagation 1.5 s, from the GOVERNED 6 mph (owner-selected, PE to confirm) and the 7 mph overspeed trip; loaded train. 10 mph shown for information only (hardware capability, not permitted)

Hand calculations

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
B1-a	Emergency braking force vs wet adhesion of braked tare (no slide)	F = 0.12·ΣW_tare (562 kN)	F/ (0.15·ΣW_tare)	67.472	84.34	kN	0.80	OK	
B1-b	Emergency stop from governed 6 mph on 2% down, loaded - distance vs 20 m criterion (ASSUMED)	a = F/m - g·0.02 = 0.632 m/s²	d = v·t + v²/2a	9.719	20	m	0.49	OK	
B1-c	Emergency stop from the 7 mph overspeed trip on 2% down	a 0.632 m/s²	d = v·t + v²/2a	12.446	30	m	0.41	OK	30 m criterion ASSUMED. Information only: from 10 mph (hardware capability, not permitted) 22.5 m
B1-d	Service stop from governed 6 mph on 2% down (70% force) vs 30 m	a = 0.383 m/s²	d = v·t + v²/2a	13.409	30	m	0.45	OK	
B1-e	Parking: all spring brakes hold loaded train on 2%, FS 1.5	required 1.5·0.02·W	F_avail = ΣF_spring	23.981	67.472	kN	0.35	OK	
B1-f	Parking on loco spring brakes ALONE (train) - not permitted	loco 4 discs		23.981	20.604	kN	1.16	FIXED by rule - train parks with all spring brakes + chocks	NG → procedure: train parks with BP exhausted (all cars spring-applied) + chocks; loco alone may hold only itself

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
B1-g	Loco AXLE disc (L3, Y+ side away from the reducer) - clamp needed PER CALIPER with 2 calipers per disc (Ø340, r_eff 135, μ_pad 0.35) vs spec 22 kN	T_axle 3.73 kN·m (1:1 - no gear multiplication)	N = T/(2 calipers · 2 μ r)	19.759	22.0	kN	0.90	OK	Reference: Coremo Ocmea D 3.5N spring-applied air-released, 24.2 kN clamp (supplier §5) - AWAIT brake supplier. Burnish pads at commissioning (initial torque 30-50 % low until bedded)
B1-l	Caliper release air: regulated supply vs Coremo D 3.5N max release pressure	MR 8.5 bar → regulator	spec	6.0	6.0	bar	1.00	OK	Dedicated regulated release-air supply off MR (6 bar max for Coremo; 5.1 bar min) - supplier F9
B1-h	Coach axle disc caliper clamp needed (Ø300 disc, r_eff 125) vs spec 10 kN	T_axle 0.71 kN·m	N = T/(2 μ r)	8.11	10.0	kN	0.81	OK	Spring-applied caliper ≥ 10 kN clamp (AWAIT supplier)
B1-i	Loco disc temperature rise, one emergency stop on 2%	E 449 kJ, share 0.076	ΔT = E·share/(m_disc c)	4.178	150	K	0.03	OK	
B1-j	Drag braking 100 m of 2% at governed 6 mph - loco disc ΔT	(0.02-0.005)·W·100 m	ΔT = E·share/(m c)	11.168	150	K	0.07	OK	Unit case; real route descents in B1-j2 (2% is the GOVERNING design grade - the track is NOT level)
B1-j2	Drag braking the longest continuous route descent (redline Alt A-D: 7.68 m fall over 505 m, avg 1.52%) at governed 6 mph - loco disc ΔT, NO cooling credit	E = W·(Δz - 0.005·L)	ΔT = E·share/(m c)	38.382	150	K	0.26	OK	Adiabatic upper bound (≈ 3.2 min at 6 mph, disc cooling ignored). Hydrostatic/engine braking credit would reduce it - AWAIT. Ops rule: ≤ 6 mph on descents
B1-k	Emergency in-train buff at loco-tender drawbar vs 70 kN service	trailing mass × a - trailing brake force		4.104	70	kN	0.06	OK	
B1-m	MS05 T04 parking brakes (4 motors through 3:1) hold the LOCO ALONE on 2%, FS 1.5	1.5·0.02·W_loco	F = ΣT04·3/r	5.865	52.4	kN	0.11	OK	Loco alone only - a train parks on all spring brakes (B1-e/f)

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
B1-n	MS05 T04 dynamic (emergency) energy per motor, loco-only stop from 6 mph vs catalog limit	max 10 dynamic applications (Poclain)	E = ½mv²/4	17.9	85.9	kJ	0.21	OK	Release by charge pressure via a PLUS+1 solenoid + accumulator: applied only on park or < 0.5 mph (engine stall does NOT dump them)
B1-o	Loco brake ratio matched to the coaches (draft-gear run-in, K1-a3): loco emergency decel vs coach decel	loco 0.690 / tender TB-3.13d 5.845 kN = 0.671 / coaches 0.69 m/s²	spec	0.69	0.69	m/s²	1.00	OK	REQUIRED (Stress Bot R3): loco SERVICE and emergency brake rate ≤ 0.69 m/s² (= loaded coaches) - caliper clamp / relay valve set at commissioning; brake test records the loco-alone decel
B1-p	Fail-safe brake logic (BP 5.0 bar UIC / Knorr KE + WABCO IR-2 971 899 152 0 inverting relay + ASCO 8210G035 24 VDC de-energize-to-vent): loss of BP, MR or 24 V → spring apply	4 failure paths on B-601	spec	1	1	logic	1.00	OK	PE to verify PL of the s30 → ASCO / CCO path; AWAIT WABCO IR-2 curve vs Coremo release 5.1-6 bar

Materials

• Discs: loco Ø340×25 on the axle (Y+, free side), 2 calipers each, coach/tender Ø300×20, cast iron GG25/ductile, or 4140 (ASSUMED)
• Pads sintered, μ 0.35-0.40 (ASSUMED; AWAIT supplier)
• Loco caliper reference Coremo Ocmea D 3.5N (25.4 mm disc thickness check AWAIT); release air via 6 bar regulator off MR; MS05 T04 multidisc brakes = loco-only parking + motor emergency (B1-m/n)
• Air: 3 × Midland Haldex 19820 main reservoirs (47 L, 150 psi) on the tender X3600-4440 (TB-3); BP 5.0 bar, MR cut-out 8.5 bar, safety valve 10 bar; Ingersoll Rand 2340 compressor with electric clutch (tender)
• Towing: wind-off calipers + T04 release screws + H1P053 tow bypass, max 2 mph (tag-out) [SUPERSEDED: tow ≤ 1 mph, E11-S5 ruling 6]

Bolts and torques

• Caliper mounts M20 cl.10.9, 610 N·m (dry K 0.20), Loctite 243 + witness mark

Fixes logged

• Tare-based brake ratio 12% adopted (loaded-only sizing slid empty coaches on wet rail)
• L3: loco brake back on an axle disc on the free side (Y+); axle torque is 1:1 so 2 spring calipers per disc (B1-g)
• Parking rule: never rely on loco brakes alone for a train (B1-f)
• r_eff 145 → 135 mm per real calipers: clamp 17.5 → 18.8 kN, 22 kN spec still holds (supplier F8)
• Commissioning: pad burnish/bedding step before brake tests; regulated release-air supply (supplier F8/F9)
• L4: MS05 T04 parking/emergency brakes (B1-m/n), independent overspeed relay to BP exhaust, tow release procedure, loco brake ratio matched to coaches (B1-o)

6 Couplers, drawbars and end restraint

Governing codes and standards

• AISC 360-22 (ASD) pins, plates, net section; AWS D1.1
• ASTM F2291 §6 (secondary restraint of coupled vehicles); NACM chain WLL

Load cases and basis

• Service 70 kN (≥ 1.9 × max adhesion-limited start 36.96 kN)
• Ultimate 200 kN (≈ 2.6 × train weight × 0.1 g impact - ASSUMED)
• Curving R100 lateral components

Hand calculations

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
C5	Loco rear drop box PL16 A572-50 root bending, 70 kN	Lever 600 mm, box 552×400 mm	see PF-CT_R2_calc_notes (calcs.py)	8.9	207.0	MPa	0.04	OK	
C5	Loco rear drop box PL16 A572-50 root bending, 200 kN	Lever 600 mm, box 552×400 mm	see PF-CT_R2_calc_notes (calcs.py)	25.6	345.0	MPa	0.07	OK	
C5w	Loco rear box root weld, 10 mm fillet all round, 200 kN	E70 weld metal: 0.6×FEXX = 289 MPa nominal (ultimate case)	see PF-CT_R2_calc_notes (calcs.py)	67.5	289.2	MPa	0.23	OK	
C5	Tender drop box PL16 A572-50 root bending, 70 kN	Lever 286 mm, box 702×400 mm	see PF-CT_R2_calc_notes (calcs.py)	3.0	207.0	MPa	0.01	OK	
C5	Tender drop box PL16 A572-50 root bending, 200 kN	Lever 286 mm, box 702×400 mm	see PF-CT_R2_calc_notes (calcs.py)	8.7	345.0	MPa	0.03	OK	
C5w	Tender box root weld, 10 mm fillet all round, 200 kN	E70 weld metal: 0.6×FEXX = 289 MPa nominal (ultimate case)	see PF-CT_R2_calc_notes (calcs.py)	31.0	289.2	MPa	0.11	OK	
C6a	Drawbar PL32×150 A572-50 net tension at Ø61 hole, 200 kN	Net section (150−61)×32	see PF-CT_R2_calc_notes (calcs.py)	70.2	345.0	MPa	0.20	OK	
C6b	Drawbar bearing on Ø60 pin, 200 kN	Bearing limit 1.5 Fy (ultimate case)	see PF-CT_R2_calc_notes (calcs.py)	104.2	517.5	MPa	0.20	OK	
C6c	Drawbar 1473 mm buckling (weak axis, pinned) at 70 kN buff	Euler Pcr=373 kN; require ≥2.0× service	see PF-CT_R2_calc_notes (calcs.py)	140,000.0	372,635.5	N	0.38	OK	Pins allow rotation in plan; vertical rotation restrained by pocket plates - pinned assumed (conservative).

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
C6c	Drawbar 800 mm buckling (weak axis, pinned) at 70 kN buff	Euler Pcr=1263 kN; require ≥2.0× service	see PF-CT_R2_calc_notes (calcs.py)	140,000.0	1,263,309.4	N	0.11	OK	Pins allow rotation in plan; vertical rotation restrained by pocket plates - pinned assumed (conservative).
C6d	Pin Ø60 4140 Q&T bending, 200 kN, spacer bushings (4 mm gap)	Double shear pin; M=F/2×(gap+t_plate/2+t_bar/4)	see PF-CT_R2_calc_notes (calcs.py)	103.7	655.0	MPa	0.16	OK	
C6e	Pin Ø60 double shear, 200 kN		see PF-CT_R2_calc_notes (calcs.py)	35.4	393.0	MPa	0.09	OK	
C6f	Pocket plates PL20 bearing on pin, 200 kN		see PF-CT_R2_calc_notes (calcs.py)	83.3	517.5	MPa	0.16	OK	
C6g	Safety chains 2× Grade 100 1/2 in, WLL vs 70 kN service	NACM WLL 15,000 lbf each (verify with chain supplier certificate)	see PF-CT_R2_calc_notes (calcs.py)	70,000	133,400	N	0.52	OK	
C6h	Safety chains 2× Grade 100 1/2 in, MBS vs 200 kN	Min. break = 4×WLL per NACM grade 100	see PF-CT_R2_calc_notes (calcs.py)	200,000	533,600	N	0.37	OK	
K1-a	Coach drawbar angle on R100 WITH ≥10 m spirals (steady) - empty coach leading-wheel L/V = curving Y/Q (Heumann) + drawbar lateral at max adhesion TE 37.9 kN	angle (5180+736.5)/R = 3.39° (old K1-a 1.82° was 1.86× low); Q empty 9.86 kN; Y/Q curving 0.861	L/V = Y/Q + F·sinθ·(1+o/s)/(2Q)	1.025	1.342	L/V	0.76	OK	Nadal 70°, μ 0.3 (1.34) - quasi-static; gauge-face lubrication recommended
K1-a2	Same, Alt A as drawn (tangent-arc, NO spirals): transient drawbar angle from the 3-D sweep	angle 7.30° (1.9-2.1× steady)	L/V = Y/Q + F·sinθ·(1+o/s)/(2Q)	1.214	1.342	L/V	0.91	OK	Passes at the 37.9 kN adhesion limit; 70 kN service buff would give 1.51 (NG) - see K1-a3/K1-d: 10 m spirals recommended on every curve

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
K1-a3	In-train buff/draft that keeps the empty-coach wheel at Nadal on Alt A as drawn (R100, no spirals) vs max force with PRELOADED draft gear (k 20 kN/mm, 1.0 mm TOTAL play at the renew limit)	allowable 51.7 kN (tender 38.5); 2 mm slack @ 50 kN/mm gives 67.4/65.6 kN	6-mass lumped model (calcs/track/drawgear.py)	35.3	38.5	kN	0.92	OK	REQUIRED (Stress Bot R3): preloaded draft gear, renew at 1.0 mm TOTAL play (F770 inspection), k ≥ 20 kN/mm (≤ 100 on the R80 route, K1-h); loco brake rate ≤ 0.69 m/s² (B1-o)
K1-b	Pin retention: head + Ø10 cross-bolt + R-clip vs 1 g up-lift of drawbar (end restraint)	drawbar 1473 PL32×150 = 55 kg	τ cross-bolt	20.602	256.0	MPa	0.08	OK	3 g vertical
K1-c	Max service drawbar force (adhesion-limited start) vs 70 kN design	μ 0.25 dry on drivers		37.11	70	kN	0.53	OK	
K1-d	Drawgear slack: start snatch / emergency run-in with 2 mm slack per joint (k 50 kN/mm, steel pins) vs 70 kN service	6-mass model; loco 1.03 vs cars 0.69 m/s²; brake 0.3 s + 0.25 s/car		67.4	70	kN	0.96	OK	1 mm at 200 kN/mm already gives 79-88 kN; 10 mm @ 50 gives 158 kN (Stress Bot 206-235 with 48 kN HPRV TE). Matching the loco brake ratio only trims run-in 10-15 %: slack control governs
K1-e	DB-800 loco-tender (kept as drawn): exact-geometry swing at R60 - tender pocket (300 mouth)	loco side 19.32° vs 32.4°	2-D pin geometry (drawbar_L4.py)	10.98	19.2	deg	0.57	OK	PASS; transient on spirals: route L4 (K2-r)
K1-f	Drawbar pin Ø60 length under head vs grip 108 + washer + cross-bolt hole + end distance (L-104 drawn 250 corrected)	grip 108 + 2 + 15 + 15		140	150	mm	0.93	OK	Specify 150 under head (170 o/a); 12 pins (10 + 2 spares)

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
K1-h	Draft-gear basis (Stress Bot R3): 1.0 mm TOTAL play at the renew limit, k swept 20-200 kN/mm, loco brake ≤ 0.69 m/s² - peak start-snatch / run-in force at k 100 vs route F (R80) coach buff allowance	k 20: 35.3 kN; k 100: 41.3 kN; k 150: 50.1 kN (NG)	6-mass model (calcs/track/drawgear_S3.py)	41.3	45.8	kN	0.90	OK	Specify k 20-100 kN/mm for R80 (route E R70 allowance 38.8 kN limits k to ≤ 25). Stiffer is NOT safer: k 150-200 gives 50-55 kN
K1-g	Safety chain 5/8 in G100 WLL vs 70 kN single-chain service; shackle Crosby G-2130 1 in WLL 83.4 kN	2 per joint		70	83.4	kN	0.84	OK	Chain WLL 100.5 kN; shackle governs

Materials

• Drawbars PL32×150 A572-50; pins Ø60 4140 Q&T, H8/f7; bushes 4140
• Safety chains 2 × 5/8 in Grade 100 per joint (WLL 100.5 kN each) with Crosby G-2130 1 in bolt shackles (WLL 8.5 t)
• 5 joints: DB-800 loco-tender, DB-1473 elsewhere; 12 pins Ø60 × 150 under head (10 + 2 spares)

Welds (AWS D1.1, E70XX, prequalified WPS unless noted)

• Pocket plates PL20 to box: 10 mm fillet all round, MT

Bolts and torques

• Cross-bolt M10 cl.8.8 + nyloc, 50 N·m (dry K 0.20)

Fixes logged

• F11: safety chains added
• Spacer bushings limit pin gap to 4 mm (C6d) - SUPERSEDED 28 Oct: 4 mm pin gap gives > 70 kN snatch/run-in (K1-d); basis now (Stress Bot R3) 1.0 mm TOTAL play renew limit, preloaded draft gear k ≥ 20 kN/mm (≤ 100 for R80, K1-h)
• K1-a corrected 28 Oct 2026: drawbar angle (5180+736.5)/R (was (2437+736.5)/R, 1.86× low) + curve-entry transient from the 3-D sweep; L/V now includes the curving guidance force (Heumann) and uses Nadal 1.34 (μ 0.3), not 0.8

7 Tender T2 - structure, engine/pump/genset mounts, fuel, cooling

Governing codes and standards

- AISC 360-22; AWS D1.1 / D1.3 (sheet)
- NFPA 30A / 49 CFR 393.65-.67 (fuel tank practice, reference); NFPA 37 (stationary engines, reference)
- EPA Tier 4 engine (AIE V3307) installation per AIE/Kubota manual (AWAIT)

Load cases and basis

- Equipment restraint 3 g long. / 2 g vert. / 1 g lat. (ASSUMED, same as ballast)
- Ambient 43 °C (110 °F) cooling design

Hand calculations

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
E-AIE	AIE V3307 package + radiator + H1P053 mount bolts 2×M16 cl.8.8 per isolator: combined T+V (AISC J3.7)	610 kg, CG 500 above mounts, mount spacing 900 long.; Fx 3 g = 17.9 kN	$(ft/Ft)^2+(fv/Fv)^2 \le 1$; Ft 0.75Fu/2, Fv 0.45Fu/2 (threads incl.)	0.008	1.0	interaction	0.01	OK	Isolator rated ≥ 3.5 kN uplift / 4.5 kN shear - AWAIT isolator supplier
E-Gens	Genset PTI-38 (TB-3 layout approved; X1060-2768) mount bolts 2×M16 cl.8.8 per isolator: combined T+V (AISC J3.7)	699 kg, CG 550 above mounts, mount spacing 1300 long.; Fx 3 g = 20.6 kN	$(ft/Ft)^2+(fv/Fv)^2 \le 1$; Ft 0.75Fu/2, Fv 0.45Fu/2 (threads incl.)	0.009	1.0	interaction	0.01	OK	Isolator rated ≥ 2.6 kN uplift / 5.1 kN shear - AWAIT isolator supplier
E-rail	Tender main rail HSS10x4x1/4 between pivots (2,600), all equipment ×2.0	w = 10.40 N/mm per rail	$M = wL^2/8 \times 2.0$	68.803	345	MPa	0.20	OK	
E-fuel	Fuel tank 150 L strapped: 2 straps 50×6 A36 at 3 g	tank 40 kg + 126 kg diesel	$\sigma = 3mg/(2 \cdot A)$	8.14	150	MPa	0.05	OK	
E-air	Tender cooling air: engine (45.6) + genset engine (~42 kW) jacket heat at ΔT_air 15 K, louvre free area at 3 m/s vs provided 2×0.9 m²	$\dot{V} = Q/(\rho\, cp\, \Delta T) = 5.05$ m³/s	$A = \dot{V} / 3$ m/s	1.683	1.8	m²	0.94	OK	Louvres 0.9 m² free area per side (intake one side, discharge other) - specified
E-run	Fuel endurance: 150 L vs 5 h evening (Tender Bot duty 14.6 L/h)	full-power screen 19.1 L/h	$t = V/q$	73.0	135.0	L	0.54	OK	90% usable

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
E-cg	Tender TB-3.13d (8,706.5 kg, CG [2422.7, 11.3, 1110.4], Tender Bot) - max wheel load vs the 26 kN rail basis	front truck share 53.0 %	lever about the pivots X1200 / 3800; ΔQ = Q·y/980	11.442	26.0	kN	0.44	OK	L4.2: TB-3.13d (+59.9 kg vs TB-3.11) supersedes TB-3.11 8,646.6 / TB-3.9 8,616.3 / TB-3.8 8,568.8
E-xh	Under-deck transformer cradle hangers (TB-3.4): 4 × L3x3x1/4, each 2 × M16 8.8 × ~140 through both HSS10x4 walls at mid-web Z963 with DOM 25.4 × 3.05 crush tubes (no welds on the rails; nearest bolt 81 mm from the X2449 crossmember weld ≥ 3d 48); whole 168.7 kg assembly at 3 g vertical + 1 g	V = 168.7·9.81·4 / 8 bolts	bolt shear, threads in the plane	0.827	30.144	kN	0.03	OK	Tender Bot: ~330 N/bolt on the 89.6 kg isolated mass at 3 g; torque M16 8.8 with crush tubes per the frame standard (153 N·m lubricated)

Materials

• Rails HSS10x4x1/4 A500C; bolsters fish-belly PL10/12 A36; deck 6 mm tread plate A786
• Skin 10 ga A1011; posts HSS2x2x1/8; fuel tank 3 mm 5052 Al or 12 ga steel with baffles, vent, rollover check valve

Welds (AWS D1.1, E70XX, prequalified WPS unless noted)

• Engine/genset mount pedestals PL16 to rails: 8 mm fillet all round
• Deck tread plate stitch 6 mm 50 @ 300 to bearers (sealed seam with sealant)

Bolts and torques

• Isolator bolts M16 cl.8.8, 219 N·m (dry K 0.20), Nord-Lock
• Fuel tank straps M12 cl.8.8, 88 N·m (dry K 0.20)

Fixes logged

• Pump H1P045 → H1P053 (§4) - same SAE B pad (AWAIT Danfoss/AIE fit)
• Cooling louvres sized (E-air)
• L4 (Tender Bot): hydraulic cooler ≥ 14 kW on the LEFT wall X3455-4145; rev TB-3 layout APPROVED 28 Oct: genset X1060-2768, R-wall NEMA 3R cabinet X2820-3780 (listed UL 1008 ATS with integral main; separate QOM22225NRB main, handle ≤ Z1639), 150 L custom tank X3900-4500 behind X3800 firewall, ODX-AGM34, 3 × Haldex MR X3600-4440; tender 8,706.5 kg (TB-3.13d, CG [2422.7, 11.3, 1110.4]; was TB-3.11 8,646.6, TB-3.8 8,568.8, TB-3.4 8,465.3) incl. the 168.7 kg under-deck shore transformer assembly (Acme T2535173S 15 kVA NEMA 3R on isolators, L3x3x1/4 cradle hung from the HSS rails by 2 × M16 8.8 through-bolts per hanger at mid-web Z963 with crush tubes - no welds on the rails, 3 mm shield 450 × 1,120 at Z300, inrush limiter on the cradle X2360-2665; nothing under the deck aft of X2900); tender CG + axle loads rerun on the real T2 CG (E-cg)
• Genset PTI-38 (699 kg) in the approved TB-3 reservation (≤ 1,775 long). Rated 40 °C, NOT the 43 °C design day; NO UL 2200 listing shown - derate letter + listing OPEN

8 Generator sizing and selection recommendation

Governing codes and standards

• NFPA 70 (NEC) 2023 Art. 445 (generators), 250.35 (separately derived), 522/525 (amusement attractions)
• ISO 8528-1 rating definitions (COP/PRP)

Load cases and basis

• Plan basis (tender-lt45 / systems-p1): electrical demand 26.8 kW hot dry evening, 31.7 kW hot dry day, 39.7 kW hot humid (110 °F / 75 °F, Δw 0.006, +15 %, COP 2.5, incl. 4.288 kW aux)
• Outside air per ASHRAE 62.1-2022 Table 6-1 'transportation waiting' analog (Rp 3.8 L/s·person, Ra 0.3 L/s·m², cabin 20.5 m², Ez 1.0 low-level supply): regular coach 33 occ. → 131.5 L/s (279 cfm), accessible 14 occ. → 59.3 L/s (126 cfm); train 454 L/s vs plan 847.5 L/s (7.5 L/s·person)
• OA load at basis conditions 41.48 W thermal per L/s = 19.08 W electric per L/s (×1.15 / COP 2.5); 62.1 OA sensible 10.7 + latent 8.2 kW thermal
• ERV: RenewAire HE05IN (120 V, 120-375 cfm, AHRI 1060 core L62-G5) one per coach; total effectiveness 0.55 cooling ASSUMED - AWAIT RenewAire selection printout; fans ≤ 343 W vs 250 W plan allowance (+0.27 kW train)
• Superseded: rev. A §8 used COP 3.0 and ΔT 11.1 K with 60 % effectiveness - inconsistent with the 39.7 kW plan basis; corrected here to the plan basis

Hand calculations

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
G1-a	Hot-humid demand WITHOUT fix (plan OA 847.5 L/s) vs 35 kW continuous	plan 39.7 kW		39.7	35.0	kW	1.13	FIXED - ERV + staging (G1-b)	NG - the 4.7 kW shortfall
G1-e	ERV total effectiveness REQUIRED to reach 34.4 kW (plan OA 847.5 L/s, incl. +0.27 kW fan delta) vs selected 0.55 ASSUMED	OA electric share 16.17 kW	ε = (D0 + ΔPfan - 34.4)/P_OA	0.344	0.55	-	0.62	OK	COP 3 sensitivity ε ≥ 0.393; at 62.1 OA ε ≥ 0 (OA cut alone reaches 32.19 kW)
G1-b	Hot-humid demand WITH HE05IN ERVs (ε 0.55), plan OA 847.5 L/s - governing	ERV saves 8.89 kW electric; fans +0.27 kW	D = 39.7 - ε·Q·19.08 W/(L/s) + ΔPfan	31.08	34.4	kW	0.90	OK	
G1-f	Hot-humid demand WITH ERVs at 62.1 OA (454 L/s) - design flow	no-ERV 32.19 kW; ERV saves 4.76 kW	same	27.7	34.4	kW	0.81	OK	
G1-c	Largest step load (staged starts: inverter heat pumps soft-start, 1 coach at a time) vs 35 kW	running 31.08 + one inverter ramp 1.5 kW (ASSUMED)		32.58	38.5	kW	0.85	OK	Gen-set transient capability 110% (ISO 8528 PRP) - AWAIT genset data

Design statements

• Recommendation: keep the 35 kW CONTINUOUS (ISO 8528 COP, not standby) single-phase 120/240 V 60 Hz Kubota-powered enclosed set that fits the rev TB-3 tender reservation (X1060–2768, i.e. 1,708 long, allow up to 1,775 × 800 × 1,000, ≤ 900 kg; layout basis PowerTech PTI-38 1,708 × 667 × 864, 699 kg), derated for 43 °C at the site, AND close the hot-humid gap with one RenewAire HE05IN ERV per coach + a demand/staging controller (shed 1 coach compressor stage if total > 33 kW).

• ERV location: a low clerestory-style roof pod on each coach (unit 880 × 956 × 307 mm, 43 kg), top ≤ 3,450 mm above rail (coach roof peak 3,089; locomotive roof 3,700; K-401 overhead 4,300). The R2 coach has no ceiling plenum (finished ceiling ≈ roof) and the end-hood pocket (1,217 W × 544 D × 431 H mm above the door header) is too shallow for any 300-cfm unit found. OA louvre and EA louvre on opposite sides/ends of the pod (62.1 Table 5-1 separation - PE to confirm).
• Alternates (data verified this turn, see calcs/erv/erv_selection.md): Mitsubishi Lossnay LGH-F300RVX2-E (208/230 V, 300 cfm, 331 mm tall, 34 kg - rejected: intake limited to 40 °C / 80 % RH, below the 43 °C design day); Fantech ATMO 300E-EC (120 V, 301 cfm, 546 mm tall - too tall for a low pod); Panasonic FV-10VEC2 (120 V, 100 cfm, 225 mm tall - 3 needed per regular coach). No 12/24 V ERV of this airflow was found.
• Alternative if ERVs are rejected: 45 kW continuous set (needs bay re-check - the LT45.1 gaps 124/94/25 mm will not take a larger enclosure without re-layout).
• Tender layout rev TB-3 (Tender Bot design_basis §11, APPROVED 28 Oct 2026): genset X1060–2768 Y417–1084; right-wall outward-facing NEMA 3R electrical cabinet X2820–3780 (panel QO142M225PRB, factory-enclosed listed 200 A NEMA 3R UL 1008 ATS with integral main, charger SENS S2); main breaker in a SEPARATE QOM22225NRB enclosure with its handle at Z ≤ 1,639 above top of rail (NEC 404.8 met without a step); custom 150 L 5052 Al fuel tank X3900–4500 behind a steel firewall at X3800; genset start battery ODX-AGM34 X2850–3200; 3 × Haldex 19820 main reservoirs X3600–4440. Tender ≈ 8,465.3 kg (Tender Bot TB-3.4, Steering 16): under-deck shore transformer ASSEMBLY 168.7 kg at CG [2502, 24, 608] between the frame rails - Acme T2535173S 15 kVA UL 506 NEMA 3R upright (Z420–851) on 4 elastomer isolators, L3x3x1/4 cradle (cross angles X2280–2356 / X2644–2720) hung from the HSS rails by 2 × M16 8.8 through-bolts per hanger at mid-web Z963 with DOM crush tubes (no welds on the rails, ≥ 3d from any weld), bolted 3 mm ballast/spray shield 450 × 1,120 at Z300, inrush limiter on the cradle box rails X2360–2665 (nothing under the deck aft of X2900); placed equipment 2,051.9 kg at CG ≈ [2221, +41, 1480]. Tender CG on the real T2 structure CG: [2433, +10, 1109] (E-cg). Transformer top vs X2500 crossmember: CLASH 15 mm (K2-xf).
• Genset purchase is NOT released (spec approval first) - procurement spec above; AWAIT supplier quotes.

Fixes logged

• 4.7 kW shortfall closed by HE05IN ERV + staging (G1-b 31.1 kW at plan OA; 27.7 kW at 62.1 OA) - owner/PE decision on ERV vs 45 kW set
• §8 rev. A COP 3.0 / ΔT 11.1 K inconsistency corrected to plan basis (COP 2.5, ΔT 19.4 K)

9 Coaches R2 - frame, body, roof, floor, doors, stairs, seating, restraint, egress

Governing codes and standards

- AISC 360-22; AWS D1.1/D1.3
- ASTM F2291-23 §6 (patron containment, restraint analysis), F770 (operations, evacuation), F1193
- IBC 2021 §1607 (100 psf assembly floor, 50 plf / 200 lb guards), §1011 stairs; NFPA 101 §7 egress (reference)
- 2010 ADA Standards §234/1002 amusement rides - accessible coach (PE/owner)

Load cases and basis

- Floor 100 psf (4.79 kPa) crowd + 300 lb point; dead 0.5 kPa
- Seats: 3 g longitudinal on seat + occupants (ASSUMED; F2291 accel. envelope ≤ 0.2 g at the governed 6 mph)
- Roof 20 psf maint. + 300 lb point; wind 100 mph parked on walls (ASCE 7-22 basic, ASSUMED)
- Guards 50 plf / 200 lb at 42 in

Hand calculations

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
C1	Coach main rail HSS12x4x1/4 bending, vertical ×1.3	Sprung loaded coach 9476 kg uniform on 2 rails over 10160 mm, bolsters at 2337/7823 mm; M=12.5 kN·m/rail at ×1.0	see PF-CT_R2_calc_notes (calcs.py)	48.0	207.0	MPa	0.23	OK	×1.3 service vs 0.6Fy; ×2.0 overload vs Fy. Side sills and floor ignored (conservative).
C1	Coach main rail HSS12x4x1/4 bending, vertical ×2.0	Sprung loaded coach 9476 kg uniform on 2 rails over 10160 mm, bolsters at 2337/7823 mm; M=12.5 kN·m/rail at ×1.0	see PF-CT_R2_calc_notes (calcs.py)	73.8	345.0	MPa	0.21	OK	×1.3 service vs 0.6Fy; ×2.0 overload vs Fy. Side sills and floor ignored (conservative).
C1d	Coach end-overhang deflection (×1.0)	Cantilever 2337 mm; limit L/360	see PF-CT_R2_calc_notes (calcs.py)	1.7	6.5	mm	0.25	OK	
C1b	Coach main rail, vertical ×1.3 + 70 kN buff (eccentric)	Coupler 550 mm vs rail centroid 635 mm: e=85 mm	see PF-CT_R2_calc_notes (calcs.py)	64.2	207.0	MPa	0.31	OK	
C2	Coach bolster HSS12x4x1/4 + 2×PL10×100 doublers, centre load ×1.3	P=46.5 kN at centre plate, span 1728 mm (rail c/c), simply supported	see PF-CT_R2_calc_notes (calcs.py)	76.7	207.0	MPa	0.37	OK	Doublers PL10×100 top+bottom over centre 600 mm; internal PL10 diaphragm each side of centre plate.
C2	Coach bolster HSS12x4x1/4 + 2×PL10×100 doublers, centre load ×2.0	P=46.5 kN at centre plate, span 1728 mm (rail c/c), simply supported	see PF-CT_R2_calc_notes (calcs.py)	118.0	345.0	MPa	0.34	OK	Doublers PL10×100 top+bottom over centre 600 mm; internal PL10 diaphragm each side of centre plate.

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
C10d	Coach platform guard post 1-1/2 in sch 40 A53B, 200 lb at 42 in	Cantilever post from floor/side sill	see PF-CT_R2_calc_notes (calcs.py)	177.8	144.6	MPa	1.23	FIXED - 2 in sch 40 posts	
C10d	Coach platform guard post 2 in sch 40 A53B, 200 lb at 42 in	Cantilever post from floor/side sill	see PF-CT_R2_calc_notes (calcs.py)	103.3	144.6	MPa	0.71	OK	
C15a	Stair tread formed channel 3/16 in (4.8) x 280 go x 40 flanges, 914 span, 300 lb point	Simply supported between stringers; IBC 1607 300 lb concentrated on 4 in² (ASSUMED code basis)	see PF-CT_R2_calc_notes (calcs.py)	70.4	150.0	MPa	0.47	OK	
C15b	Stair tread deflection, 300 lb point	Limit L/240	see PF-CT_R2_calc_notes (calcs.py)	0.7	3.8	mm	0.19	OK	
C15c	Stair stringer PL8x150 A36, 100 psf crowd + dead (per stringer)	Horizontal span 1550 mm hinge-to-foot; plan area 1.28 m²; 100 psf = 4.79 kPa (IBC 1607 stairs, ASSUMED)	see PF-CT_R2_calc_notes (calcs.py)	23.6	150.0	MPa	0.16	OK	
C15d	Stair stringer deflection (along slope)	Limit L/360	see PF-CT_R2_calc_notes (calcs.py)	0.7	5.2	mm	0.13	OK	
C15e	Chain stay 3/8 in Grade 70 (x2), foot unsupported, full crowd	Moment of full load about the floor hinge / 548 mm chain lever arm; WLL 6,600 lbf for 3/8 G70 ASSUMED - verify on chain tag / supplier (AWAIT)	see PF-CT_R2_calc_notes (calcs.py)	5.2	29.4	kN	0.18	OK	
C15f	Floor hinge pin Ø25 4140 Q&T (2 per stair), single shear	Hinge reaction with chains loaded; brackets/bolts to side sill by PE	see PF-CT_R2_calc_notes (calcs.py)	9.4	262.0	MPa	0.04	OK	
C15g	Stair handrail 1-1/2 in sch 40 A53B, 200 lb at mid-span (662 along slope)	Posts at treads 1, 3, 5; IBC 1607.9 200 lb any direction (ASSUMED)	see PF-CT_R2_calc_notes (calcs.py)	27.6	144.6	MPa	0.19	OK	
C15h	Stair handrail post 2 in sch 40 A53B, 200 lb at 895 above tread	Fold-flat post locked upright by strut + Ø16 pin	see PF-CT_R2_calc_notes (calcs.py)	86.6	144.6	MPa	0.60	OK	

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
C15i	Stair handrail post lock pin Ø16 4140, double shear	Couple over 100 mm hinge-to-pin	see PF-CT_R2_calc_notes (calcs.py)	19.8	262.0	MPa	0.08	OK	
R1-a	Floor crossmember HSS6x2x3/16 between rails (1,728) at 900 c/c, 100 psf + DL	w 4.76 N/mm	σ = wL²/8S	29.138	207.0	MPa	0.14	OK	
R1-b	Floor crossmember deflection, live only	L/360	δ = 5wL⁴/384EI	0.538	4.8	mm	0.11	OK	
R1-c	Outrigger HSS6x2x3/16 cantilever 533 to side sill, floor + wall/roof line 2.0 N/mm		M = w a²/2 + P a	26.82	207.0	MPa	0.13	OK	
R1-d	Floor panel (1 in composite) required stiffness per mm width for L/360 at 100 psf, span 900	spec value to supplier	EI ≥ 5qL⁴·360/(384 L)	16.368	16.368	N·m²/mm·1e-3	1.00	OK	Spec row - panel supplier to certify EI and 300 lb point (AWAIT)
R1-e	Roof bow HSS2x2x1/8 @880 (at each window post), span 2,794, deflection L/180	20 psf + 0.25 kPa dead	δ = 5wL⁴/384EI	19.518	15.522	mm	1.26	FIXED - HSS3x2x1/8 bows (R1-f)	NG → FIX: HSS3x2x1/8 roof bows
R1-f	Roof bow HSS3x2x1/8 @880 (at each window post), span 2,794, deflection L/180	20 psf + 0.25 kPa dead	δ = 5wL⁴/384EI	7.366	15.522	mm	0.47	OK	ADOPTED
R1-g	Window post HSS2x2x1/8 @880, 100 mph parked wind (Cp 0.8) + 50 plf lean, span 1,100	q 980 Pa	M = wL²/8 + P_lean·L/4	36.035	207.0	MPa	0.17	OK	
R1-h	Bench anchor (4-seat bench, 2 rear anchors) M12 cl.8.8 tension at 3 g	Fx 10.24 kN, CG 600, base 450	T = Fx·h/(b·2) - DL/4; Ft = 0.75Fu/2·As	5.972	25.29	kN	0.24	OK	Through-bolt to crossmember or 100×100×6 backing plate - no inserts in composite floor
R1-i	Egress width: NFPA 101 0.2 in/person (level) for 32 pax vs 2 × 813 end doors		W_req = 0.2 in × n	162.56	1,626	mm	0.10	OK	

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
R1-j	Evacuation time to ground: stair deploy 60 s + 32 pax at 0.5 p/s per stair × 2 stairs vs 180 s (ASSUMED target)		t = t_deploy + n/(2·0.5)	92.0	180	s	0.51	OK	Train crew procedure per ASTM F770; drill to verify

Design statements

• Occupant restraint (ASTM F2291 restraint analysis): accelerations at the governed 6 mph (owner-selected limit, PE to confirm) with 1.0 m/s² emergency are ≤ 0.11 g longitudinal, ≤ 0.04 g lateral on R100 (quasi-static lateral scales with v²: 0.06 × (6/7.5)²) → seated patrons need seat backs + handholds; no individual restraint; gates latched and interlocked to traction (no movement with a gate open).
• Doors/gates: end doors 32 in (813) clear; platform side gates 42 in high, latched + proximity-switch interlock to the traction-enable loop (fail-safe).
• Emergency-exit stairs both sides at every door (C15, C-304); stowed latches with proximity switches in the same interlock loop.

Materials

• Underframe: HSS12x4x1/4 rails, HSS6x2x3/16 crossmembers/outriggers/side sills A500C
• Body: HSS2x2x1/8 posts, 16 ga skin, roof bows HSS3x2x1/8 @880 (fix), polycarbonate glazing 6 mm (FMVSS 205 AS-? AWAIT)
• Floor: 1 in composite panel, fire/smoke per NFPA 130 Table A practice (reference)

Welds (AWS D1.1, E70XX, prequalified WPS unless noted)

• Crossmembers/outriggers to rails: 5 mm fillet all round
• Window posts to side sill and eave: 3 mm fillet all round (D1.3 for 16 ga skin plug welds)

Bolts and torques

• Bench anchors M12 cl.8.8, 88 N·m (dry K 0.20), nyloc, backing plate
• Stair hinge pins Ø25 4140 + R-clip; handrail posts lock pin Ø16

Fixes logged

• Roof bows HSS2x2x1/8 @880 deflect 1.12×limit → HSS3x2x1/8 (R1-e/f)
• Guard post 1-1/2 → 2 in sch 40 (C10d)
• Seat anchors through-bolted (R1-h)

10 Electrical (train supply) and coach HVAC

Governing codes and standards

• NFPA 70 (NEC) 2023 Art. 400 (flexible cords), 445, 250, 522/525; NFPA 101 §7.9 emergency lighting (90 min)
• UL 1995 / AHRI 210/240 HVAC equipment; ASHRAE 62.1 ventilation (reference)

Hand calculations

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
H1-a	Genset main breaker 175 A vs NEC 445.12(A) max OCPD (rev 2026-11-02, per Electro electrical_plan)	genset rated 158.3 A (≈38 kW / 240 V 1-ph); 35 kW cap 145.8 A	OCPD ≤ 1.15·I_rated = 182 A (445.12(A)); 175 A ≤ 182 A, 200 A not allowed; load util 145.8/175	145.8	175	A	0.83	OK	basis = PTI-38 integral 175 A output breaker (SELECTED, not tested); fallback QOM2175VH in QOM22225NRB. Old "125%" / "OCPD ≥ 1.15·I" wording was wrong.
H1-b	First train-line jumper (all 4 coaches) type W 2/0 3C+G (ampacity 195 A, NEC 400.5 col. B, ASSUMED)	34.4 kW target (≥ G1-b 31.1 kW) incl. ERV case	I = P/V	179.167	195	A	0.92	OK	Continuous load ×1.25
H1-c	Coach branch #4 SOOW (70 A) for 10 kW coach peak		I = P/V ×1.25	52.083	70	A	0.74	OK	
H1-d	Voltage drop 65 m train line 2/0 Cu at 143 A, 1-phase	R 0.33 Ω/km (2/0 Cu)	ΔV = 2·I·R·L	2.556	3.0	%	0.85	OK	

Design statements

• HVAC: one inverter heat-pump (mini-split class) per coach + one RenewAire HE05IN ERV per coach (enthalpy core, ε 0.55 ASSUMED - AWAIT printout) in a roof clerestory pod supplying the ASHRAE 62.1 outside-air rate (279 cfm regular / 126 cfm accessible) through low-level side supply; MERV 13 filters; condensate to track drain.
• Staging/demand controller in the tender AC panel: limits total to 33 kW, sheds compressor stages coach-by-coach, keeps ventilation and emergency lights.
• Grounding: genset neutral bonded at the genset (separately derived); equipment grounding conductor through every jumper; GFCI on 120 V receptacles; bonding jumpers across trucks.
• Emergency lighting: battery-backed 90 min per coach; exit/orientation 10 W per coach (plan EL1).

Fixes logged

• 4.7 kW hot-humid shortfall closed (ERV + staging) - see §8

11 Wind - 40 mph no-run, overturning, tie-downs

Governing codes and standards

• ASCE 7-22 Ch. 26-29 (parked, basic wind speed - site value AWAIT from ASCE hazard tool; 100 mph ASSUMED)
• ASTM F2291 §6 environmental loads; operating limit per F770

Hand calculations

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
C11	Coach empty wind overturning FS at 67 mph gust	Side area 23.2 m², Cd 1.2, centroid Z1750 mm, restoring m·g·gauge/2	see PF-CT_R2_calc_notes (calcs.py)	1.5	1.4	FS req/avail	1.09	FIXED by rule - 40 mph no-run + tie-downs (C11w, W1)	Requirement FS ≥ 1.5 ASSUMED; engineer to set per ASTM F2291 environmental loads.
C11	Coach loaded wind overturning FS at 67 mph gust	Side area 23.2 m², Cd 1.2, centroid Z1750 mm, restoring m·g·gauge/2	see PF-CT_R2_calc_notes (calcs.py)	1.5	2.3	FS req/avail	0.67	OK	Requirement FS ≥ 1.5 ASSUMED; engineer to set per ASTM F2291 environmental loads.
C11	Locomotive wind overturning FS at 67 mph gust	Side area 35.0 m², Cd 1.2, centroid Z1900 mm, restoring m·g·gauge/2	see PF-CT_R2_calc_notes (calcs.py)	1.5	1.8	FS req/avail	0.86	OK	Requirement FS ≥ 1.5 ASSUMED; engineer to set per ASTM F2291 environmental loads.
C11	Tender wind overturning FS at 67 mph gust	Side area 9.5 m², Cd 1.2, centroid Z1430 mm, restoring m·g·gauge/2	see PF-CT_R2_calc_notes (calcs.py)	1.5	4.2	FS req/avail	0.36	OK	Requirement FS ≥ 1.5 ASSUMED; engineer to set per ASTM F2291 environmental loads.
C11w	Coach empty overturning FS at 40 mph gust (adopted no-run limit)	Owner decision R2: no running above 40 mph gusts; parked train tied down (wind tie-downs)	see PF-CT_R2_calc_notes (calcs.py)	1.5	3.9	FS req/avail	0.39	OK	
C11w	Coach loaded overturning FS at 40 mph gust (adopted no-run limit)	Owner decision R2: no running above 40 mph gusts; parked train tied down (wind tie-downs)	see PF-CT_R2_calc_notes (calcs.py)	1.5	6.3	FS req/avail	0.24	OK	
C11w	Locomotive overturning FS at 40 mph gust (adopted no-run limit)	Owner decision R2: no running above 40 mph gusts; parked train tied down (wind tie-downs)	see PF-CT_R2_calc_notes (calcs.py)	1.5	4.9	FS req/avail	0.30	OK	
C11w	Tender overturning FS at 40 mph gust (adopted no-run limit)	Owner decision R2: no running above 40 mph gusts; parked train tied down (wind tie-downs)	see PF-CT_R2_calc_notes (calcs.py)	1.5	11.7	FS req/avail	0.13	OK	

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
W1-Coac	Coach empty parked 100 mph: tie-down force per windward tie (2 ties) vs G70 3/8 chain WLL 29.4 kN	F 34.1 kN at Z1750; 0.6D restoring	T = (M_o - 0.6 M_r)/(gauge·2)	20.803	29.4	kN	0.71	OK	
W1-Loco	Locomotive parked 100 mph: tie-down force per windward tie (2 ties) vs G70 3/8 chain WLL 29.4 kN	F 51.4 kN at Z1900; 0.6D restoring	T = (M_o - 0.6 M_r)/(gauge·2)	24.123	29.4	kN	0.82	OK	
W1-Tend	Tender parked 100 mph: tie-down force per windward tie (2 ties) vs G70 3/8 chain WLL 29.4 kN	F 14.0 kN at Z1430; 0.6D restoring	T = (M_o - 0.6 M_r)/(gauge·2)	0.0	29.4	kN	0.00	OK	

Design statements

- No-run above 40 mph gust (anemometer at the station, ops rule per F770); all vehicles FS ≥ 3.8 at 40 mph (C11w).
- Parked: 2 ties per vehicle each side (4 per vehicle) to helical anchors / deadman - anchor capacity ≥ 30 kN each AWAIT geotech.

12 Track / rail interface - wheel profile, gauge, curve clearances

Governing codes and standards

• ASTM F2291 §6 (clearance envelope); AREMA Ch. 4/5 practice (reference); ASCE 40 lb rail section properties

Hand calculations

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
K2-a	Max static driver wheel load (L4.1: Loco Bot ledger 01:57 - barrel 2,076.2 + saddles 195.7, smokebox 227.9 incl. SBX-7, fittings 40.3, deck +36.3, cab C1 813.9, BT-1 + 2 × 75.4 kg plates + clamp bars in the front pocket, slats 21.9, HL-1 headlamp 25, TC/valve package 28, hub 31.4/axle, no divider, sprung pony) vs revised rail basis 26 kN			25.89	26.0	kN	1.00	OK	Rail basis raised 24 → 26 kN (ASCE 40 base stress ≈ 101×26/22.85 = 115 MPa vs 172 allow.)
FT-1a	Loco front tow lug FT-1 (L-111): 8 × Lindapter HB16-2 in the front end sill, 70 kN at 20°/15°, worst bolt T 14.0 / V 3.7 kN	axial 63.5 kN; shear/bolt 3.67 kN	T/21 + V/30 (HB16 SWL basis)	0.79	1.0	-	0.79	OK	Loco Bot value verified; HB16-2 clamp range to confirm with Lindapter; holes drilled, ≥ 50 mm from weld toes
FT-1b	FT-1 end-sill wall yield line (HSS12x6x3/8, t 8.86, panel 278 × 328 clamped, patch 126 × 236) - Loco Bot Pu 220 kN / 1.67	independent pyramid Pu 280.6 kN (0.9× = 252.5)	70 / (220/1.67)	0.53	1.0	-	0.53	OK	governing lower bound kept; independent 0.46
FT-1h	FT-1 Crosby G-209 shackle (Loco Bot)			0.84	1.0	-	0.84	OK	governing FT-1 item
K2-a2	Loco Bot S17: equal-torque dry 2 % start margin after the front-ballast trim (1,056 -> 150.8 kg plates + BT-1 68.2 kg in the pocket) - rule >= 1.10	lightest axle F2 37.11 kN	rule / margin	1.1	1.14	-	0.96	OK	smokebox 227.9 kg counted in full (no double count: the L3 smokebox front left the base with the ~1,000 kg L3 cab/front)

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
K2-hb	Loco Bot S17: S4N saddle holes at bearer B4 (HSS5x3x3/16, front bolster) Y±250 - net section with the REQUIRED fix (M12 hex rivnuts on the bearer CL, pairs ±25 in Y)	M 7.47 kN·m (LC4P 2 g)	f / (0.6·Fu·Afn/Afg), AISC F13.1	136.1	194.08	MPa	0.70	OK	as proposed (2 rivnuts at X±25 in one section): 1.03 NG, and the hex-16 hole runs 5.6 mm into the 9.5 corner radius (rivnut cannot seat) - applies to every F-07 saddle rivnut S1-S6; B1/S1 0.15, B4 Y±480 0.41
K2-hs	Loco Bot S17: S1 smokebox saddle rivnut tension (227.9 kg, 1 g roll + 1 g up) and pilot slat holes (20 × M10 in the L3 PL12 face, 6 % row loss)		T / SWL 4 kN (ASSUMED)	0.991	4.0	kN	0.25	OK	pilot slats OK with conditions: >= 30 mm from box edges/welds, clear of the coupler pocket Y±560 and the FT-1 load path, full 12 mm thread or through-bolted
K2-ss	Deck side stringers SS± C3x4.1 -> HSS3x3x1/8 (P-A1, ADOPTED S17), grillage with S17 loads		M/S vs 0.6 Fy	81.31	207.0	MPa	0.39	OK	was 0.57 on C3x4.1; bearer max 0.67
K2-r70	L-109 R70 swept half-width (Loco Bot sweep, K-401 method): cab eave governs	R100 1,581.6 reproduced	envelope vs 1,696	1,629.4	1,696.0	mm	0.96	OK	66.6 mm margin CONFIRMED; front-end items 198 / 547 / 559
K2-b	ASCE 40 rail base stress at 26 kN (scaled from 22.85 kN screening) vs 172 MPa		σ ∝ Q	114.923	172	MPa	0.67	OK	
K2-c	Rigid wheelbase 1,550 on R100 - versine vs flangeway play 12.4		f = wb²/8R	3.003	12.4	mm	0.24	OK	
K2-d	Loco swept half-envelope on R100 - 3-D kinematic sweep of the R2 model (governing point: CAB ROOF eave, incl. 2° sway + 6.2 play) vs K-401 half 1,696	old single-point cab-step check 1487 (understated - wrong governing point; K-401 L2 and the sweep agree within 4 mm at R45: 1,719 / 1,715)	route_check.py sweep	1,581.603	1,696	mm	0.93	OK	R50 1,691.7 (+widening 1,701 NG → cab-roof eave trim ≥ 10 mm if R50 is adopted); R45 1,715 NG

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
K2-d2	Coach swept half-envelope on R100 incl. Trim Bot garland/lights 60 mm per side vs K-401 1,696	coach 1586.5 + 60	sweep + 60	1,646.529	1,696	mm	0.97	OK	R50 1,688 (+widening 917.5 → 1,693 OK; 920.7 → 1,697 NG: keep R50 widening at 917.5 or garland ≤ 50 mm)
K2-e	Driver top to running-board edge angle (underside Z1,530): vertical clearance - 25 bump vs 50 min			50	55	mm	0.91	OK	
K2-f	Driver tyre sweep under running board at R100 (truck yaw) - lateral movement at wheel end vs board overhang beyond tyre 703	yaw 0.92°	Δy = 1,500·sin(yaw)	24.0	703	mm	0.03	OK	
K2-g	L4 chain case (lowest part, inside the frame) vs rail top - case bottom Z389	case bottom Z389, rail top Z0	vertical clearance - 25 bump - 50 wheel wear	75	314	mm	0.24	OK	
K2-h	Decorative cylinders Ø380 on T1 at Y±790 (L4, moved from Y±750) vs lead-truck wheel rim - min gap after 6.2 play on Alt A as drawn (R100, no spirals) vs 10 mm	at Y±750 it was 0.1 (NG)	sweep	10	37.013	mm	0.27	OK	FIXED rev L4: cylinders at Y±790 (25 mm at R50)
K2-i	Loco-tender DB-800: tender-side pocket swing (300 mouth, limit 19.2°) - max angle on Alt A as drawn	loco side 13.3° vs 32.4°	sweep	10.016	19.218	deg	0.52	OK	R50 needs 19.5-19.7° → tender pocket mouth 150 (limit 32.4°) if R50 is adopted
K2-c2	Rigid 1,550 driver truck on R45 - versine incl. flange lead vs flangeway play at gauge -3 (9.4)	2-axle truck cannot jam; versine criterion with flange lead	f = (wb+2·lead)²/8R	7.62	9.4	mm	0.81	OK	Stress Bot's 13.2 mm (= wb²/4R) is a 3-axle criterion - not applicable to 2-axle trucks

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
K2-j	Max gauge accepted by the wheelsets: widened 920.7 + 6 build + 6 wear (932.8) - tread overlap on the rail head (worn flange 17, min B2B 857, tread 120, chamfer 5)	overlap 56.2 vs ASCE 40 head 47.6		932.8	941.4	mm	0.99	OK	Absolute max 941.4; maintenance limit 932 proposed
K2-k	Restraining rail flangeway, min case (widened 917.5, check gauge 886 +2/-0) vs 25 mm (22 flange + 3)	flangeway = G - C		25.0	26.6	mm	0.94	OK	No widening (914.4) → 23.4 NG: restraining rail only with widened gauge
K2-b2	ASCE 40 rail base stress, track_structure spec (ties 6×8×6 ft @ 22.5 in, 2×26 kN) vs 172 MPa	track engineer calc reproduced		153	172	MPa	0.89	OK	Lateral basis 0.35×2×26 = 18.2 kN (engineer used 17.1). Their 'L/V 0.11 at R100' is centrifugal only - flange guidance gives Y/Q ≈ 0.8-0.9 on R≤100: design curves < R100 for 0.5×axle
K2-l	Lead (pony) truck, ACTUAL wheel load 2.9 kN (LT42: body reaction 0; PE 5.4 was wrong) - free swing arm, leading wheel Y/Q on R45 vs Nadal 1.34	Q 2.89 kN (2.15 in twist); Heumann 1,050 wb	Y/Q chord mode	0.942	1.342	L/V	0.70	OK	Passes ONLY with no lateral centering force: allowable arm/centering friction 0.4 kN at Q_min (μ 0.3, 1.2 margin). REQUIRED: load the pony (K2-m) + swing stops ±120 mm (Stress Bot R3; R80 working route swing 89.5 mm, E 102, A 72, D 119)

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
K2-m	Pony load 15 kN (upper end of 10-15) taken as a BODY SPRING (reacted at X1,525, outside the pivots) - max driver wheel vs 26 kN	spring lifts 1.80×F off the front truck: F2 axle 11.67 kN, rear wheel 26.5	statics	26.5	26.0	kN	1.02	REJECTED OPTION - sprung pony with own ballast adopted (K2-m2)	Body-spring loading over-loads the rear truck and strips the front drivers (equal-torque cap 25.2 → 13.9 kN). RECOMMENDED: sprung pony carrying its OWN 1.25 t ballast (12.3 kN, Q 5.96 kN/wheel, no effect on the drivers)
K2-m2	ADOPTED L4: sprung pony carrying its OWN 1.25 t ballast (no load transfer to the drivers) - max driver wheel vs 26 kN	K2-m body-spring option REJECTED	statics	25.89	26.0	kN	1.00	OK	Pony Q ≈ 5.96 kN/wheel → K2-l lead-wheel Y/Q improves; swing stops ±120 mm
K2-n	Barrel saddles S3 (X2450) / S5 (X5675), bearers and RB-1 brackets vs the L4 chain drive (Loco Bot) - min plan gap	drive top Z1061 vs RB bracket bottom Z1175.0	CAD envelope	50	389	mm	0.13	OK	CLEAR; keep-out zones on L-103
K2-lo	K-401 LOWER OUTLINE (Steering 14b): all non-wheel parts >= 150 mm ATOR between the rails / >= 100 mm outside; level track + vertical curves R >= 500 m (crest/sag loss included), new wheels, static	governing: transom HSS10x4x1/4 (upright, pair) tender clear 196.0 mm (between); loco/tender/coach checked from the CAD model	limit / clearance	150.0	196.0	mm	0.77	OK	PASS all vehicles. Not included: wheel wear + spring deflection (tender/coach transom margin 46 mm; PE to set wheel-wear limit) and coach stairs (deployed only at rest). Tender transformer shield bottom Z300 (TB-3.4, 150 margin); inrush box bottom Z380.

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
K2-fp	Fouling point / clearance mark, #8 turnouts (Rail Bot, Steering 14c): track centres at the mark vs 2 x 1,696 dynamic half-width + 200	mark at 3.6 m centres; #8 frog 7.153 deg, lead R 84.9 m → 33.99 m from the points (Rail Bot 34.0 m)	2·1,696 + 200	3,592.0	3,600	mm	1.00	OK	Clearance marks 34.0 m from the points of each #8 - VERIFIED (33.99 m). Working layout F (R80, level), owner pending. If either track is on R80 at the mark, add ~37 mm per side (loco end throw 185 vs 148 at R100) - within the 200 mm allowance (74 < 200); level layout F, owner pending.
K2-xf	Tender shore transformer (TB-3.13d) top vs the X2500 HSS10x4 frame crossmember underside Z836 - cradle DROPPED 50 mm (kept since TB-3.8)	unit top Z851 → Z801 static; Z811 at the isolator stop	Z_top vs Z_underside	811	836	mm	0.97	OK	CLOSED: 25 mm at the isolator stop, 100 mm to the K-401 outline
K2-rF	Route L4 Alt F R80 - WORKING ROUTE (owner pending) (final spiralled, L4 masses): max L/V (loco/tender/empty coach at full TE 37.7 kN) vs Nadal 1.34	env loco 1609.6 / coach+garland 1657.1 vs 1,696; ΔQ/Q 0.29 vs 0.6	Heumann Y/Q + F·sinθ·fac/2Q	1.27	1.342	L/V	0.95	OK	PASS
K2-rE	Route L4 Alt E R70 (final spiralled, L4 masses): max L/V (loco/tender/empty coach at full TE 37.7 kN) vs Nadal 1.34	env loco 1635.5 / coach+garland 1670.7 vs 1,696; ΔQ/Q 0.33 vs 0.6	Heumann Y/Q + F·sinθ·fac/2Q	1.32	1.342	L/V	0.98	OK	PASS
K2-rA	Route L4 Alt A R100 (final spiralled, L4 masses): max L/V (loco/tender/empty coach at full TE 37.7 kN) vs Nadal 1.34	env loco 1581.7 / coach+garland 1646.5 vs 1,696; ΔQ/Q 0.23 vs 0.6	Heumann Y/Q + F·sinθ·fac/2Q	1.18	1.342	L/V	0.88	OK	PASS

ID	Check	Inputs	Formula	Demand	Capacity	Unit	Util	Status	Note
K2-rD	Route Alt D R60 with truck-pitch transfer at the traction pin, HIGH-RAIL GAUGE-FACE LUBRICATED (Nadal 1.84 at flange μ 0.15) - lubricators on ALL R ≤ 80	dry L/V 1.381 vs 1.342 (NG dry); F 1.258, E 1.309	Heumann Y/Q + F·sinθ·fac/2Q	1.381	1.84	L/V	0.75	OK	CLOSED (L4.1): top of rail kept dry for 2 % starts; lubricator maintenance in the operating manual

Design statements

• Wheel profile (all vehicles): 1:20 tread taper, flange height 25 mm, flange thickness 22 mm at gauge point (10 mm below tread), flange angle 70°, root radius 12, tread width 120.
• Back-to-back 858 ±1.0 mm (G 914.4 - 2×22 - 2×6.2 play). WHEELSET flange-face gauge 902 nom / 891 min worn (857 + 2×17); wheel check dimension (B2B + one flange) 881 max new / 874 min worn. (The former 'check gauge 892 min' was the worn wheelset gauge, a WHEEL dimension - it is NOT a track check gauge: as one it leaves a 22 mm flangeway, NG.)
• Track gauge 914.4 +6/-3 (tangent, R ≥ 75); widened 917.5 (36-1/8 in) on R50-75 and 920.7 (36-1/4 in) on R45-50 - accepted by all wheelsets (K2-j). Absolute max gauge 941.4 (full head under worn tread); MAINTENANCE LIMIT 932 (renew/regauge).
• Restraining (guard) rail on curves < R60: TRACK CHECK GAUGE (high-rail gauge face to restraining face) 886 +2/-0, renew below 883 (must stay > 881 wheel check dimension); FLANGEWAY = G - 886 = 31.6 at 917.5 / 34.8 at 920.7 (min 26.6 / 29.8 ≥ 25). Restraining rail REQUIRES the widened gauge (at 914.4 the flangeway is 28.4, min 23.4 < 25 NG). The track engineer's 886 is correct; the PE '892' is the wheel dimension above.
• Design grade 2% GOVERNING (route redline 28 Oct 2026; curves compensated so grade + curve resistance ≤ 2.0%). The track is NOT designed level; 'laser-levelled 0%' reports do not change the basis until the owner confirms.
• Minimum radius: design basis 100 m (unchanged). R60/R50 study results and the changes each needs: calcs/track/route_check.md.
• [SUPERSEDED: rail RULED 60 lb bolted relay rail - section checks to be re-run] Rail ASCE 40 lb/yd new (track_structure spec), ties 6×8 in × 6 ft at 22.5 in (572), tie plates, 9/16 in cut spikes - consistent with the 26 kN wheel basis (K2-b2). Curves < R100: lateral design 0.5×axle, rail braces or 4 spikes/plate on the high rail, gauge-face lubrication. Ballast ≥ 10 in under ties (Stress Bot subgrade 26-34 psi at 6 in vs 20).

A Fastener torque schedule (ISO 898-1; preload 75% proof; mark after torque)

Size	Class	As mm²	Preload kN	Torque dry/zinc K0.20 N·m	Torque lubricated K0.14 N·m
M10	8.8	58.0	25.2	50	35
M10	10.9	58.0	36.1	72	51
M12	8.8	84.3	36.7	88	62
M12	10.9	84.3	52.5	126	88
M16	8.8	157.0	68.3	219	153
M16	10.9	157.0	97.7	313	219
M20	8.8	245.0	110.2	441	309
M20	10.9	245.0	152.5	610	427
M24	8.8	353.0	158.8	762	534
M24	10.9	353.0	219.7	1055	738

Running-gear and drive fasteners: Nord-Lock wedge washers or prevailing-torque nuts + witness mark; re-torque after 10 h running. Structural slip-critical joints (if any added): ASTM F3125 A325 per RCSC 2020 turn-of-nut.

B Weld schedule

Joint	Weld	Size	Inspection
Drop drawbar box to rails / sill	Fillet all round	10	100% VT + MT
Pocket plates PL20 to box	Fillet all round	10	100% VT + MT
Loco crossmembers to main rails	Fillet all round	6	VT
Bolster box seams	Fillet both sides	6	VT
Centre plate to bolster	Fillet all round	8	VT + MT
Powered truck horn cheeks to chords	CJP groove + 6 fillet cap, toes ground	full pen.	VT + UT 100%
Transom box seams	Fillet both sides	8	VT + MT 10%
MS05 motor bracket PL25 to transom (L4)	2 × 400 fillet both sides + 2 PL16 gussets	10	VT + MT
Chain oil-bath case seams (3 mm)	Continuous fillet, leak test	3	VT + leak
Engine/genset pedestals to tender rails	Fillet all round	8	VT
Coach crossmembers/outriggers to rails	Fillet all round	5	VT
Window posts / roof bows	Fillet all round	3	VT
Stair stringers / treads (C-304)	Fillet both sides	5	VT

All welding by AWS D1.1-qualified welders to a written WPS; CWI inspection of primary members (drawgear, trucks, drive brackets). Pre-heat per D1.1 Table 5.8 for A572-50 > 20 mm.

C Fix log

#	Subsystem	As-found problem	Fix	Ref
1	Drive	Owner plan direct drive / L1 1:1 on Ø1,450: 21.9 kN vs 31.2 kN 2% start	Rev L2 custom spur gear (1.81:1), rev L3 Dodge TA7315H15MS + H1B110 + belt, then rev L4 (owner: no belt) Poclain MS05 468/234 + ANSI 120-2 chain 3:1 - 37.7 kN at 350 bar	F25, F26, D1-a
2	Drive (L4.1)	Flow divider (L4 per-motor slip control) not buildable: Jahns HTO-4-35-FI runs below its 12-16 L/min minimum at every start, 320 bar section rating caps TE at 31.2 kN < 32.3 needed, +160 kg	Divider DELETED; electronic traction control (per-motor sensors + pony ground-speed reference, pump destroke, auto-sand, start in 1st, hill-start hold); TE cap = equal torque; dry/sanded 2% margin 1.10 kept with 2 × 75.4 kg plates	D1-c, D1-m, D1-m2 [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
3	Truck (L4.1)	Truck-pitch weight transfer missing (Stress Bot r2): pivot h 1.08 m → equal-torque 2 % start fwd 0.94 / rev 0.69	LOW TRACTION PIN per truck at h 0.65 m (Ø100 4140 pin, GE100ES under the transom, centre plate flat-lined): fwd 1.13; reverse PLUS+1 pull cap 28.9 kN + reverse full-train starts only ≤ 1.20 % (L4.2) grade (covers Ø1430 worn drivers); K2-a 25.7 kN with transfer	D4-wt/wr
4	Drive (L4.1)	Trip run-in: CCO/24 V trip in 1st at 400 bar gave 67 kN, accidental T04 52 kN vs 33.2 kN R80 limit	Pump ramp to neutral ≥ 3 s, motors de-energize to 2nd, T04 pressure switch vents BP first + choked T04 (≥ 1.0 s + 2.0 s): worst 23.7 kN	D4-ri
5	Drive (L4.1)	Chain slack check used dL = dC (L = C)	dL = (2·span/C)·dC; ±22 mm bump stops, ±1 mm side float, re-shim at the centre plate (primary ≤ 3 mm + motor-slide reset); 2 % elongation hard limit; B29.1 certificate ≥ 302.5 kN; zero-speed reversal	D2-f/D4-ch
6	Axle (L4.1)	Hub fit 0.12-0.17 with an oil-press option (SF 0.87 pressed)	SHRINK ONLY 0.15-0.20; hot hub 30 K + smoothing SF 1.87 at HPRV; removal groove kept	D4-hf
7	Drive (L4.1)	Kubota high idle could exceed 6 mph at the 42 cc stop	Stroke stop 38 cc sealed: 5.91 mph at 2,850 rpm high idle	D4-sp
8	Route (L4.1)	Route D R60 L/V 1.39 > 1.34 (K2-rD); with transfer E/F marginal	High-rail gauge-face lubricators on ALL R ≤ 80 (Nadal 1.84): D 1.32 margin - CLOSED	K2-rD
9	Tender (L4.1)	Transformer top Z851 vs X2500 crossmember Z836 (K2-xf)	Tender Bot TB-3.8: cradle dropped 50 mm (25 mm at the isolator stop, 100 mm to K-401) - CLOSED; tender TB-3.9 8,616.3 kg, brake 5.825 kN = 0.676 m/s² vs loco 0.690 (within 0.03)	K2-xf/E-cg
10	Loco body (L4.1)	Loco Bot: saddle web windows cut the 3 g longitudinal path (util 5.9); B1 hole X460 in the sill weld zone; SD-1 hole outside the cap; SD-2 hold-down open; clamp bars for 1 plate/side	8 gussets per saddle (0.28); B1 holes X480/600; SD-1 cap re-cut to Y±50 (0.84); SD-2 clamp plates under the sill flange (no holes); clamp bars 0.59 at 30 N·m	L41-sw/s1/s2/cb

#	Subsystem	As-found problem	Fix	Ref
11	T04 release (L4.1 s7)	T04 release held by a 0.75 L accumulator on the brake line: an engine stall left the motor brakes released ~30 s (loss of charge did NOT apply them); a 4th dry-break was proposed	Release supply from the loco loop (LP shuttle + 20 bar reducing/relieving valve + normally-vented solenoid); choke Ø2.4 + 1.0 L damping accumulator (delay 1.01 s, rise 2.97 s); manual release from a 6.0 L accumulator + hand pump via a guarded valve; loss of loop/charge or 24 V applies the T04 after the BP vents; no 4th dry-break	L41-t1/t2
12	Wheels (L4.1 s7)	No driver condemning limit; pony Ø630 worn gives a 35.0 mm flange vs ~32.5 mm to the ASCE 40 joint-bolt heads	Driver condemning DIAMETER Ø1430 adopted (6 mph governor / 7 mph trip, adhesion 1.116, clearances pass); FLANGE-HEIGHT limit 28 mm / ≤ 27 (L4.2 FINAL) on all loco wheels (re-profile); reverse-start rule ≤ 1.20 % (L4.2); diameters entered in PLUS+1 at gauging	L41-w1..w5
13	Brake air (L4.1 s8/9)	Loco release time not specified; Loco Bot estimate 3.7 s at an assumed C 2.0 (range 1.8-7.3 s); release regulator/relay not sized for flow	Target ≤ 4 s WORST CASE; release path total C ≥ 3.0 (required 2.5) → 3.46 s incl. 1.0 s signal; parts budget C 5.26 → 2.4 s: relieving regulator C ≥ 10, 1/2 in relay valve C ≥ 8 piloted by the IR-2, 3/4 in trunk C ≥ 8, 4 × 1/2 in branches C ≥ 2.0, quick-exhausts, release-proof switch; timed commissioning test	L41-b1..b4
14	Drive (L4.1)	AIE SAE B 13T pump drive mates only H1P053 shaft G4 (222 N·m max < 265 engine peak)	Keep G1; engine coupling hub 14T 12/24 (AIE confirmation held)	D1-u
15	Drive (L4.1)	15T spline engagement 70 mm ASSUMED, code-5 spline only ~60 mm	Full-length engagement ≥ 58 mm, 95 MPa (0.95); 4140 15T if shorter	D2-h
16	Axle (L4.1)	Ring bolts 10.9 fitted (non-catalog) and Ø340 flange edge 20 mm = 1.18 d0	Catalog 8.8 fitted DIN 609/7968, 219 N·m; Ø350 flange edge 25 mm = 1.47 d0; hub 31.4 kg	X1-g, X1-h
17	Drive (L4)	Equal torque limited by the lightest axle (L4 early pass F2 29.6 kN): 2% start margin 0.92	(L4: flow divider) → sand mandatory wet; front ballast trimmed to 150.8 kg (2 × 75.4 plates) + BT-1 68.2 kg in the pocket (Steering 17, C9 rule) → lightest axle 35.63 kN, dry equal-torque margin 1.10	D1-a/c/m [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
18	Drive (L4)	Chain case at 3% stretch + ±25 travel had < 15 mm radial clearance	Case enlarged (axle-end inner r 332.7)	D2-g
19	Drive (L4)	Source Bot 56 links would put the motor 38 mm into the transom	54 links (C 417.6 mm new); slotted mount + 2 M20 jack screws, no idler	D2-e/f
20	Axle (L4)	Ø140 pressed seat n 1.37 without hardening credit	Ø150 raised seat, no key: n 1.64 on the heaviest axle	X1-c
21	Drive (L4)	Overspeed trip ran through PLUS+1	Independent 2-channel speed relay to BP exhaust + pump neutral	D1-n
22	Drive (L3)	H1B110 in line on the TA II input does not fit (rear Y491 vs sideframe 330 / wheel 427; hits horn cheek)	1:1 synchronous-belt drop: motor through the open sideframe window, 6 mm inside the wheel line	F27, D1-p
23	Axle (L3)	Ø120 keyed seat, 4140 profile keyseat: fatigue n 0.96 < 1.5	34CrNiMo6 Q&T + sled-runner keyseat + induction-hardened seat: n 1.82	X1-c/d, C7s

#	Subsystem	As-found problem	Fix	Ref
24	Drive (L3)	One spinning wheelset could drive the TA II input to 2,106 rpm (> 1,790)	Per-motor speed sensors + PLUS+1 anti-slip destroke	D1-l/m
25	Brakes (L3)	Axle disc is now 1:1 - one caliper would need ~35 kN	2 spring calipers per disc, 17.5 kN each vs 22 kN spec	B1-g
26	Drive	DT5 hub key 243 MPa at relief; KTR FA25 at 1.10×	Eliminated (L2/L3: outboard stub/coupling removed)	F03
27	Drive	Pump H1P045 flow limited top speed	H1P053 kept; L4: 6 mph governed by the sealed 38 cc stop (5.91 mph at 2,850 high idle; 5.4 at rated); 7.6 mph hardware max, independent 7 mph trip	D1-f
28	Drive	Pump input 350 N·m at 380 bar > G4 shaft 222 N·m	G1 input shaft 534 N·m (368 N·m at 400 bar) + anti-stall software 35 kW (supplier F2)	D1-e
29	Drive	Relief 380 bar not a Danfoss setting	Option L35: pressure limiter 350 / HPRV 400 bar; torques re-run (supplier F3)	X1-a, D1-g
30	Brakes	Caliper r_eff 145 optimistic	r_eff 135 → 18.8 kN clamp, 22 kN spec holds; Coremo D 3.5N reference; burnish + regulated release air	B1-g, B1-l
31	Drive	Wet-rail 2% start short 9.2 kN	Sanders on leading axle each direction (μ ≥ 0.21)	D1-c [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
32	Drive	F25-A memo: nose reaction taken as 0.75 × torque (wrong)	Full axle torque to nose link (C7n)	C7n
33	Brakes	L2: axle disc space lost to the gear case	L3: disc back on the axle, free side (Y+), 2 calipers	B1-g
34	Brakes	Loaded-weight brake sizing slides empty coaches on wet rail	Tare-based 12% braking ratio	B1-a
35	Brakes	Loco spring brakes alone cannot hold the train on 2%	Park with BP exhausted (all spring brakes) + chocks	B1-e/f
36	Trucks	Coil primary springs do not fit the 170 mm LT18 window	Rubber-metal chevron springs (spec)	T1-d
37	Coaches	Roof bows HSS2x2x1/8 @880 deflect 1.26 × L/180	HSS3x2x1/8 bows	R1-e/f
38	Coaches	Guard post 1-1/2 in sch 40 overstressed	2 in sch 40	C10d
39	Loco	Ballast end wall PL12 overstressed	PL20 + 4 × M20 hold-downs	C9
40	Wind	Empty coach FS 1.37 at 67 mph	40 mph no-run + parked tie-downs	C11w, W1
41	Drive (L4.2)	Stress Bot r3: descents on hydrostatic retarding; T04 burst/accidental apply; 2,850 rpm high idle; 30 mm flange limit with 6 mm joint wear	FLOAT mode (loop bypass, ≥ 3 s swash ramp), friction brakes sized for the full 2 % descent; Ø1.2 T04 port orifices + hard lines/4SH + 2-channel vent switch, T04 fed from the H1 CHARGE circuit (loop shuttle deleted); hard tamper-sealed 2,600 rpm + 42 cc; flange 28 mm / re-profile ≤ 27, 2 mm joint wear: run-in worst 25.3 kN ≤ 33.2	L42-dn, L42-t1..t4, D4-ri, D4-sp

#	Subsystem	As-found problem	Fix	Ref
42	Pressure (L4.2)	No -16 assembly ≥ 450 bar; strength at HPRV 400 only	PL 350 / HPRV 380; loop parts ≥ 420 bar WP with ISO 18752 / ISO 6605 impulse rating covering the ~430 bar spike; every STRENGTH case at 430 bar: 15T sprocket made to order in 4140 Q&T (C1045 stock hub 1.03 NG); chain MBL SPEC ≥ 326 kN certified (standard 296 kN not accepted)	D2-h, D4-mb, D1-g
43	Mass (L4.2)	K2-a drifted to 26.1 kN (final drive 336.7 kg/axle, gross brake piping/chains, Ø120 pins, P-106 pockets)	NB-1 nose ballast 59.8 kg at X450 (ahead of the front pivot unloads the rear truck) + PLUS+1 forward TE command limit 33.0 kN: K2-a 25.89 kN of record; reverse pull cap 28.9 kN	L42-k2
44	Power	Hot-humid coach HVAC 4.7 kW over 35 kW	RenewAire HE05IN ERV per coach (roof pod) + staging controller (31.1 kW at plan OA)	G1-b/e/f
45	Track	Rail basis 22.85 kN exceeded (24.5 kN with L3)	Rail basis 26 kN, ASCE 40 min.	K2-a/b

D Revision history

Rev	Date	Description
R1/R2	2026-10-27	Design review, calc notes C1-C16, 13-sheet set
L1/L1b	2026-10-27	Loco to owner plan Ø1,450 drivers; 4 motors
L2	2026-10-27	Inboard spur-gear drive; full PE-review calc package
L3	2026-10-28	Owner: choose the best fit → catalog Dodge TA7315H15MS 14.9:1 + Danfoss H1B110 via 1:1 belt (L2 custom case rejected); Ø120 seat fix; axle discs 2 calipers; governed 6 mph owner-selected (PE to confirm)
L4	2026-10-28	Owner: no belt → Poclain MS05 468/234 2-speed + T04 frame-mounted, ANSI 120-2 15T:45T 3:1 chain in oil bath, Ø150 pressed hub; H1P053 stroke-stop governor; independent overspeed; tender T2 update; Loco Bot interfaces; sprung pony; cylinders Y±790
L4.1	2026-10-28	Flow divider deleted → electronic traction control; Loco Bot ledger 01:57 + HL-1 headlamp 25 kg + TC/valve package 28 kg + clamp bars; hub Ø350 flange 31.4 kg; 8.8 fitted ring bolts 219 N·m; shaft code 5 W55x3x17 / hub N55x3x17, ≥ 58 mm engagement; pump G1 + 14T 12/24 coupling; T04 release + accumulator + tow valve specified; 15T OD 202; axlebox C ≥ 190 kN; 2 × 75.4 kg plates + clamp bars; DBK-1/jumper zones recorded (this document)
L4.2	2026-10-28	Stress Bot r3 rulings: float-mode descents, T04 orifices + charge feed, 2,600 rpm sealed limit, flange 28 / ≤ 27, Ø120 pins + centre pad, sanders both trucks, reverse limit from the calc, pony 96T; PL 350 / HPRV 380, loop ≥ 420 bar, strength at 430; TB-3.13d; E7/E8 brake air (QEV B); NB-1 nose ballast, K2-a of record [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]